

Forcing and its Applications

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Any mistakes, inaccuracies, or omissions in these notes are solely my responsibility and should not be attributed to the lecturers. Please feel free to contact me at cliu4@uni-muenster.de.

Contents

0 Basic Knowledge	3
1 Boolean-Valued Forcing	6
2 Applications	15
2.1 A model of ZFC with no "definable" well-order of reals and a model of ZF in which there is a countable set of countable sets of reals whose union is uncountable.	15
2.1.1 A model of ZFC with no "definable" well-order of reals	15
2.1.2 A model of ZF in which there is a countable set of countable sets of reals whose union is uncountable	19
2.2 Solovay's model: A model of ZF where every set of real is Lebesgue measurable and has the Baire property and perfect set property	22
2.2.1 The BIG Theorem and its LONG Proof	29

0 Basic Knowledge

We begin with a bunch of definitions:

- A **forcing poset** (/forcing/forcing notion) is a triple $(\mathbb{P}, \leq_{\mathbb{P}}, \mathbb{1}_{\mathbb{P}})(=\mathbb{P})$ such that $\leq$ is a **pre-order** on the set $\mathbb{P}$, elements of $\mathbb{P}$ are called **conditions**(/forcing conditions). We say that q is stronger than p if $q \leq_{\mathbb{P}} p$.
- $D \subseteq \mathbb{P}$ is **dense** if $\forall p \in \mathbb{P}, \exists q \in D(q \leq p)$.
- $G \subseteq \mathbb{P}$ is a **filter** if $\forall p \in G, \forall q \in \mathbb{P}(p \leq q \rightarrow q \in G)$ and $\forall p \in G, \forall q \in G, (\exists r \in G, r \leq p, r \leq q)$.
- For $p, q \in \mathbb{P}$, we say p and q are **compatible** if $\exists r \in \mathbb{P}$ such that $r \leq p, r \leq q$, denoted as $p \parallel q$. Otherwise it is called **incompatible**, denoted as $p \perp q$.
- $\mathbb{P}$ is called **atomless** if $\forall p \in \mathbb{P}, \exists q, r \leq p(q \neq r \rightarrow q \perp r)$, a condition $p \in \mathbb{P}$ is called **atom** if $\forall q, r \leq p(q \neq r \rightarrow q \parallel r)$.
- $A \subseteq \mathbb{P}$ is an **antichain** if and only if $\forall p, q \in A, (p \neq q \rightarrow p \perp q)$. We say $\mathbb{P}$ has the **countable chain condition** (c.c.c.) if every antichain is countable.
- Let $\mathcal{D}$ be a set of dense subsets of $\mathbb{P}$, a filter $G \subseteq \mathbb{P}$ is called **$\mathcal{D}$ -generic** if $\forall D \in \mathcal{D}, G \cap D \neq \emptyset$.

Let V be a transitive model of ZFC, let $\mathbb{P} \in V$ be a forcing poset. A filter $G \subseteq \mathbb{P}$ is **V -generic** if it is $\mathcal{D}_V$ -generic where $\mathcal{D}_V = \{D \subseteq \mathbb{P} : D \text{ dense}\} \cap V$.

If $\mathbb{P}$ is atomless then a V -generic filter can never be an element of V .

Example 0.1. Let I, J be two non-empty sets, let $\kappa \leq \bar{I}$ be infinite cardinal. Then $\text{Fn}_{<\kappa}(I, J)$ is the set of all functions $p: \text{dom}(p) \rightarrow J$ with $\text{dom}(p) \subseteq I$ and $\overline{\text{dom}(p)} < \kappa$.

Consider the following forcing $(\mathbb{P}, \leq, \mathbb{1}) : \text{Fn}_{<\kappa}(I, J), \leq = \{(q, p) \in \mathbb{P}^2 : q \supseteq p\}, \mathbb{1} = \emptyset$.

For each $i \in I, D_i := \{p \in \mathbb{P} : i \in \text{dom}(p)\}$ is dense. Let $\mathcal{D} = \{D_i : i \in I\}$, then $\mathcal{D}$ -generic filters correspond to functions $g: I \rightarrow J, G \mapsto \bigcup G, \{p \in \mathbb{P} : p \subseteq g\} \leftrightarrow g$.

Lemma 0.2. Let $\mathbb{P}$ be a forcing poset and $\mathcal{D}$ to be a countable set of dense subsets of $\mathbb{P}$. Then for every $p \in \mathbb{P}$, there exists a $\mathcal{D}$ -generic filter $G \subseteq \mathbb{P}$ with $p \in G$.

Definition 0.3 (Forcing Axiom). Let κ be a cardinal and $\mathbb{P}$ be a forcing poset. $\text{FA}_{\kappa}(\mathbb{P})$ is the statement that whenever $\mathcal{D}$ is a set of dense subsets of $\mathbb{P}$ with $\overline{\mathcal{D}} \leq \kappa$, then there exists a $\mathcal{D}$ -generic filter $G \subseteq \mathbb{P}$.

Notice that FA_{ω} holds for every $\mathbb{P}$. $\text{FA}_{\aleph_1}$ fails for certain $\mathbb{P}$.

Example 0.4. Let $\mathbb{P} = \text{Fn}_{<\omega}(\omega, \omega_1)$, consider $D_i := \{p \in \mathbb{P} : i \in \text{dom}(p)\}$ for $i < \omega, E_{\alpha} := \{p \in \mathbb{P} : \alpha \in \text{ran}(p)\}$ for $\alpha < \omega_1$. It is clear that all D_i, E_{α} 's are dense. Let $\mathcal{D} = \{D_i : i < \omega\} \cup \{E_{\alpha} : \alpha < \omega_1\}$. $\text{FA}_{\aleph_1}(\mathbb{P})$ implies that there exists a $\mathcal{D}$ -generic filter $G \subseteq \mathbb{P}$, define $g = \bigcup G: \omega \rightarrow \omega_1$ is surjective, clearly leads to a contradiction.

Definition 0.5 (Martin's Axiom). MA_{κ} is the statement that $\text{FA}_{\kappa}(\mathbb{P})$ holds for all c.c.c. forcings.

Every countable forcing is c.c.c. $\text{Fn}_{<\omega}(I, J)$ is c.c.c. if both I and J are countable.

Note that if J is uncountable then $\text{Fn}_{<\kappa}(I, J)$ does not have the c.c.c. since fix any $i \in I$ then $\{(i, j) : j \in J\}$ is an uncountable antichain.

Lemma 0.6. *Let I be infinite then $\text{Fn}_{<\omega}(I, J)$ has the c.c.c. if and only if J is countable.*

Lemma 0.7. *If MA_κ holds then $\kappa < 2^{\aleph_0}$.*

Proof. Let $\mathbb{P} = \text{Fn}_{<\omega}(\omega, 2)$, this is a c.c.c. forcing. There exists a countable set $\mathcal{D}_0$ of dense subsets of $\mathbb{P}$ such that for every $G \subseteq \mathbb{P}$ a $\mathcal{D}_0$ -generic filter, $g = \bigcup G$ is a function $\omega \rightarrow 2$. For each $h : \omega \rightarrow 2$, define $D_h := \{p \in \mathbb{P} : p \not\subseteq h\}$. Notice that D_h is dense, if $G \subseteq \mathbb{P}$ is $\mathcal{D}_0 \cup \{D_h\}$ -generic then $\bigcup G : \omega \rightarrow 2$ is different from h . Now suppose $\text{MA}_{2^{\aleph_0}}$ holds true then there exists a filter $G \subseteq \mathbb{P}$ which is $\mathcal{D}_0 \cup \{D_h : h \in 2^\omega\}$ -generic. Then $g = \bigcup G : \omega \rightarrow 2$ but $g \neq h$ for all $h \in 2^\omega$ which clearly leads to a contradiction. ■

Lemma 0.8. *Assume MA_κ . Let $(A_\alpha : \alpha < \kappa)$ be a sequence of infinite subsets of ω such that $\forall \alpha < \beta < \kappa, A_\alpha \cap A_\beta$ is finite. Then for every $S \subseteq \kappa$, there exists $B \subseteq \omega$ such that $\forall \alpha < \kappa, B \cap A_\alpha$ is infinite if and only if $\alpha \in S$.*

Proof. Consider the following forcing $(\mathbb{P}, \leq, 1)$: $p \in \mathbb{P}$ if and only if $p : \text{dom}(p) \rightarrow 2$ with $\text{dom}(p) \subseteq \omega$ such that $\forall \alpha \in S, A_\alpha \cap \text{dom}(p)$ is finite and $p^{-1''}\{1\}$ is finite, $p \leq q$ if and only if $p \supseteq q$, $1_\mathbb{P} = \emptyset$.

To apply Martin's Axiom, we have to verify this forcing poset has the countable chain condition. Suppose there exists an uncountable antichain $A \subseteq \mathbb{P}$, since there are only countably many possibilities for $p^{-1''}\{1\}$, by pigeonhole principle, there must be $p, q \in A$ such that $p^{-1''}\{1\} = q^{-1''}\{1\}$. But then consider $r = p \cup q, r \leq p, r \leq q$ which is clearly a contradiction.

Fix $S \subseteq \kappa$, now consider the following sets:

For each $\beta \in \kappa \setminus S, D_\beta := \{q \in \mathbb{P} : A_\beta \subseteq \text{dom}(q)\}$. We claim each D_β is dense. Pick any $p \in \mathbb{P}$, define q as follows:

$$q : A_\beta \cup \text{dom}(p) \rightarrow 2, q(i) = \begin{cases} p(i), & \text{if } i \in \text{dom}(p), \\ 0, & \text{otherwise} \end{cases}$$

clearly $q \leq p, q \in D_\beta$. By $A_\alpha \cap A_\beta$ is finite for all $\alpha < \beta < \kappa$, this q is still a condition of the forcing poset.

For each $\beta \in S, k < \omega, E_{\beta,k} := \{q \in \mathbb{P} : \exists i > k, i \in A_\beta \cap \text{dom}(q) \wedge q(i) = 1\}$. We claim that each $E_{\beta,k}$ is dense. Pick $p \in \mathbb{P}$, since $\beta \in S$ so $A_\beta \cap \text{dom}(p)$ is finite, then $A_\beta \setminus \text{dom}(p)$ is infinite, so for any $k < \omega$ we can find some $i \in A_\beta$ such that $k < i < \omega$. Now we define $q \in \mathbb{P}$:

$$q : \text{dom}(p) \cup \{i\} \rightarrow 2, q(j) = \begin{cases} p(j), & \text{if } j \in \text{dom}(p), \\ 1, & \text{if } i = j. \end{cases}$$

Clearly $q \leq p, q \in E_{\beta,k}$.

Let $\mathcal{D} = \{D_\beta : \beta \in \kappa \setminus S\} \cup \{E_{\beta,k} : \beta \in S, k < \omega\}$, by MA_κ , there exists a $\mathcal{D}$ -generic filter $G \subseteq \mathbb{P}$, denote $g = \bigcup G : \text{dom}(g) \rightarrow 2$ and set $B = g^{-1''}\{1\}$.

Note that for each $\beta \in S, k < \omega, \exists i < \omega$, such that $i > k \wedge i \in A_\beta \cap B$, this shows " $\Rightarrow$ ". Also for each $\beta \in \kappa \setminus S$, there exists $p \in G$ such that $A_\beta \subseteq \text{dom}(p)$. Let $F := p^{-1}''\{1\}$, then for any $i \in A_\beta \setminus F, p(i) = 0$ since g extends p so for any $i \in A_\beta \setminus F, g(i) = 0$ thus $A_\beta \cap B$ is finite. And this shows " $\Leftarrow$ ". ■

Corollary 0.9. *If $\aleph_0 < \kappa < 2^{\aleph_0}$ then MA_κ implies $2^\kappa = 2^{\aleph_0}$.*

Proof. Consider the map $f: \mathcal{P}(\kappa) \rightarrow \mathcal{P}(\omega)$ by sending S to B . To show $2^\kappa = 2^{\aleph_0}$, it suffices to verify that f is an injection. Let S_1, S_2 be two distinct subsets of κ , $B_1 = f(S_1), B_2 = f(S_2)$ and set $S_3 = S_1 \cup S_2$. By the previous lemma, if $B_1 = B_2 = B$ then $\forall \alpha \in S_3, B \cap A_\alpha$ is infinite then leads to a contradiction of the construction of S_1, S_2 . ■

Definition 0.10. *MA is the statement that MA_κ holds for all infinite cardinal $\kappa < 2^{\aleph_0}$.*

Corollary 0.11. 1. $\text{MA} \rightarrow 2^\kappa = 2^{\aleph_0}$ for all infinite cardinals $\kappa < 2^{\aleph_0}$.

2. $\text{MA} \rightarrow 2^{\aleph_0}$ is regular (Follows from 1 and König's theorem).

- Does CH imply $\text{MA}_{\aleph_1}$? NO.
- Does CH imply MA? Trivially YES.

Theorem 0.12 (Solovay, Tennenbaum). *Assume GCH holds. Let κ be an regular cardinal greater than $\aleph_1$ then there exists a c.c.c. forcing $\mathbb{P}$ such that $\mathbb{1}_{\mathbb{P}} \Vdash_{\mathbb{P}} \text{MA} + 2^{\aleph_0} = \kappa$.*

(But we do not prove this theorem in this lecture.)

Now let $\mathcal{N}$ be the set of all Lebesgue-null subsets of $\mathbb{R}^d$, i.e. $\mathcal{N} = \{A \subseteq \mathbb{R}^d : \mu(A) = 0\}$.

Question: For what κ the following holds:

$$\text{add}_\kappa: \text{ for every sequence } (N_\alpha : \alpha < \kappa) \text{ with } N_\alpha \in \mathcal{N}, \forall \alpha < \kappa, \bigcup_{\alpha < \kappa} N_\alpha \in \mathcal{N}.$$

Theorem 0.13. *For every infinite $\kappa, \text{MA}_\kappa \rightarrow \text{add}_\kappa$.*

Observation: Under CH, add_κ fails for every uncountable κ .

Proof. Let $(N_\alpha : \alpha < \kappa)$ be a sequence of null sets in $\mathbb{R}^d$ and let $N = \bigcup_{\alpha < \kappa} N_\alpha$. To show that $\mu(N) = 0$, recall $\bar{\mu}(N) = \inf\{\mu(\mathcal{O}) : \mathcal{O} \subseteq \mathbb{R}^d \text{ open } \wedge N \subseteq \mathcal{O}\}$.

For any $\varepsilon > 0$, it suffices to find some $\mathcal{O} \subseteq \mathbb{R}^d$ open such that $N \subseteq \mathcal{O}$ and $\mu(\mathcal{O}) < \varepsilon$.

Define the following forcing poset $(\mathbb{P}, \leq, \mathbb{1})$:

$$p \in \mathbb{P} \Leftrightarrow p \subseteq \mathbb{R}^d \text{ open } \wedge \mu(p) < \varepsilon; \forall p, q \in \mathbb{P}, q \leq p \Leftrightarrow q \supseteq p; \mathbb{1} = \emptyset.$$

We now claim that $\mathbb{P}$ has c.c.c., suppose there exists a sequence $(p_\alpha : \alpha < \omega_1)$ with $p_\alpha \perp p_\beta, \forall \alpha < \beta < \omega_1$. By construction of $\mathbb{P}, p_\alpha \perp p_\beta \Rightarrow \mu(p_\alpha \cup p_\beta) \geq \varepsilon, \forall \alpha < \beta < \omega_1$.

For $A, B \subseteq \mathbb{R}^d$, define $d(A, B) = \mu(A \Delta B) = \mu(A \cup B) - \mu(A \cap B)$ and $E \subseteq \omega_1$ uncountable such that $\forall \alpha \in E, \mu(p_\alpha) \leq \varepsilon - \delta, \delta > 0$ and $d(p_\alpha, p_\beta) \geq \delta$, for all $\alpha < \beta < \omega_1 \wedge \alpha, \beta \in E$. For each $\alpha \in E$, let $q_\alpha \subseteq p_\alpha$ such that q_α is a finite union of boxes with rational coordinates and $\mu(p_\alpha \setminus q_\alpha) \leq \frac{\delta}{10}$. Then

$\forall \alpha, \beta \in E$ disjoint, $d(q_\alpha, q_\beta) \geq d(p_\alpha, p_\beta) - d(p_\alpha, q_\alpha) - d(p_\beta, q_\beta) > 0$ thus the map which sends $\alpha \in E$ to q_α is an injection as $d(q_\alpha, q_\beta) > 0$. But E is uncountable and there are only countably many q_α 's, contradiction! Thus $\mathbb{P}$ has the c.c.c.

Consider $\mathcal{D} = \{D_\alpha : \alpha < \kappa\}$ where $D_\alpha = \{q \in \mathbb{P} : N_\alpha \subseteq q\}$, it is easy to check D_α is dense. By MA_κ , let $G \subseteq \mathbb{P}$ be a $\mathcal{D}$ -filter, define $\mathcal{O}_G = \bigcup G$. Note that $\mu(\mathcal{O}_G)$ can be written as $\mu(\bigcup_{n < \omega} p_n) = \sup_{k < \omega} (\mu(\bigcup_{n \leq k} p_n)) \leq \varepsilon$ for some sequence $(p_n : n < \omega)$ with $p_n \in G$ for all $n < \omega$. And $\mathcal{O}$ is the open set we desired. $\blacksquare$

1 Boolean-Valued Forcing

Let σ_{Ba} be the first-order signature $(\wedge, \vee, \neg, 0, 1)$.

Definition 1.1. Let $S \neq \emptyset$ be a set, an **algebra over S** is a set $\mathcal{B} \subseteq \mathcal{P}(S)$ that contains S and is closed under complements and finite disjunction and conjunction.

Definition 1.2 (Or Proposition). A **Boolean algebra** is σ_{Ba} -structure $\mathbb{B} = (B, \wedge, \vee, \neg, 0, 1)$ for which there exists a set $S \neq \emptyset$ and an algebra $\mathcal{B}$ over S such that $\mathcal{B} \cong (\mathbb{B}, \cap, \cup, S \setminus \cdot, \emptyset, S)$.

Definition 1.3. A Boolean algebra $\mathbb{B}$ is a σ -structure $\mathbb{B} = (B, \wedge, \vee, \neg, 0, 1)$ which is a model of the following theory: $\forall x, y, z \in B$

$$x \vee (y \vee z) = (x \vee y) \vee z \text{ (and its dual)}$$

$$x \vee y = y \vee x \text{ (and its dual)}$$

$$x \vee 0 = x \text{ (and its dual)}$$

$$x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z) \text{ (and its dual)}$$

$$x \wedge \neg x = 0 \text{ (and its dual)}$$

$$0 \neq 1$$

Given a Boolean algebra $\mathbb{B}$, define the following relation **orderination** $\leq_{\mathbb{B}}$ on $\mathbb{B}$:

$$\forall a, b \in \mathbb{B}, a \leq_{\mathbb{B}} b \Leftrightarrow a \vee b = b \Leftrightarrow a \wedge b = a \Leftrightarrow a \wedge \neg b = 0 (\Leftrightarrow a \rightarrow b = 1)$$

Definition 1.4. $\mathbb{P} = (\mathbb{P}, \leq)$ is called a **bounded lattice** if

- $\leq$ is a partial order on $\mathbb{P}$
- $\forall a, b \in \mathbb{P}$ both $a \wedge b := \inf(\{a, b\})$ and $a \vee b := \sup(\{a, b\})$ exists.
- $\mathbb{P}$ has a least element $0_{\mathbb{P}}$ and a greatest element $1_{\mathbb{P}}$.

A bounded lattice is called **distributive** if and only if $\wedge$ and $\vee$ distribute over each other.

$\mathbb{P}$ is **complemented** if for every $p \in \mathbb{P}$ there exists $\neg p \in \mathbb{P}$ such that $p \wedge \neg p = 0_{\mathbb{P}}, p \vee \neg p = 1_{\mathbb{P}}$.

Notice that $\neg p$ may not be unique in a distributive bounded lattice if it exists.

Lemma 1.5. • If $\mathbb{B} = (B, \wedge, \vee, 0, 1)$ is a Boolean algebra then $(\mathbb{B}, \leq_{\mathbb{B}}, 0, 1)$ is a complemented distributive bounded lattice.

• If $(\mathbb{P}, \leq)$ is a complemented distributive lattice then $(P, \wedge, \vee, \neg, 0_{\mathbb{P}}, 1_{\mathbb{P}})$ is a Boolean algebra.

A Boolean algebra $\mathbb{B}$ is itself a forcing but has an atom 0, we associate to it $\mathbb{B}^+ = \mathbb{B} \setminus \{0\}$. "Forcing with $\mathbb{B}$ " refers to forcing with $\mathbb{B}^+$, e.g. $\mathbb{B}$ has the c.c.c. meant $\mathbb{B}^+$ has the c.c.c.

Definition 1.6. An **ultrafilter** on $\mathbb{B}$ is a filter $\mathcal{U} \subseteq \mathbb{B}^+$ that satisfies $\forall a \in \mathbb{B} ((a \in \mathcal{U}) \vee (\neg a \in \mathcal{U}))$.

Definition 1.7. For a topological space X , let $\text{clop}(X)$ be the set of clopen subsets of X .

Notice that $\text{clop}(X)$ is an algebra of sets on X .

Definition 1.8. A **stone space** is a topological space which is compact, Hausdorff, zero-dimensional (i.e. $\text{clop}(X)$ is a base for the topology on X)

If $\mathbb{B}$ is a Boolean algebra, $\text{St}(\mathbb{B}) := \{\mathcal{U} \subseteq \mathbb{B} : \mathcal{U} \text{ is an ultrafilter}\}$ equipped with the topology generated by the sets $N_a = \{\mathcal{U} \in \text{St}(\mathbb{B}) : a \in \mathcal{U}\}$ for $a \in \mathbb{B}$.

Now we check $\text{St}(\mathbb{B})$ is a stone space.

For Hausdorff: let U, V be two distinct point in the stone space. There is then some $a \in \mathbb{B}$ such that $((a \in U) \wedge (\neg a \in V)) \vee ((a \in V) \wedge (\neg a \in U))$ holds. Without the loss of generality we may assume that $a \in U$ and $\neg a \in V$, then $U \in N_a$ and $V \in N_{\neg a}$. Since we have that $N_a \cap N_{\neg a} = \emptyset$ thus $\text{St}(\mathbb{B})$ is Hausdorff.

For compact:

For zero-dimensional:

Theorem 1.9 (Stone's Representation theorem). (1) For every Boolean algebra $\mathbb{B}$, $\text{St}(\mathbb{B})$ is a stone space and $\mathbb{B} \xrightarrow{\sim} \text{clop}(\text{St}(\mathbb{B})), b \mapsto N_b$ and $a \leq b \Leftrightarrow N_a \subseteq N_b$.

(2) For every stone space X , $X \xrightarrow{\sim} \text{St}(\text{clop}(X)), x \mapsto \{A \in \text{clop}(X) : x \in A\}$.

Proposition 1.10. $\text{FA}_{\kappa}(\mathbb{B})$ holds if and only if the intersection of at most κ many open dense subsets of $\text{St}(\mathbb{B})$ is always non-empty.

Proof. " $\Rightarrow$ " Let $(\mathcal{O}_{\alpha} : \alpha < \kappa)$ be a sequence of open dense subsets of $\text{St}(\mathbb{B})$. Define

$$\text{For all } \alpha < \kappa, D_{\alpha} = \{a \in \mathbb{B} : N_a \subseteq \mathcal{O}_{\alpha}\}$$

Now we claim all the D_{α} s are dense. Fix some α , pick any $b(\neq 0) \in \mathbb{B}$, since $\mathcal{O}_{\alpha}$ is open dense so $\mathcal{O}_{\alpha} \cap N_b \neq \emptyset$, thus there exist some open $\mathcal{U}$ such that $\mathcal{U} \subseteq \mathcal{O}_{\alpha} \cap N_b$. Then there exists $a \in \mathbb{B}^+$ such that $\mathcal{U} \in N_a \subseteq \mathcal{O}_{\alpha} \cap N_b$, then $a \in D_{\alpha}$ and $a \leq_{\mathbb{B}} b$.

By $\text{FA}_{\kappa}(\mathbb{B})$, there exists $G \subseteq \mathbb{B}^+$, a $\{D_{\alpha} : \alpha < \kappa\}$ -generic filter, then $\{N_a : a \in G\}$ is a family of closed subsets with Finite Intersection Property. Since the space is compact, $\emptyset \neq \bigcap_{\alpha < \kappa} N_{a_{\alpha}}$ and $\bigcap_{\alpha < \kappa} N_{a_{\alpha}} \subseteq \bigcap_{\alpha < \kappa} \mathcal{O}_{\alpha}$.

" $\Leftarrow$ " Let $\mathcal{D}$ be a set of dense subsets of $\mathbb{B}^+$ with $|\mathcal{D}| < \kappa$. For every $D \in \mathcal{D}$, define $\mathcal{O}_D := \bigcup_{b \in D} N_b$, we claim all $\mathcal{O}_D$ s are open dense. Openness is clear, now we show the density: Pick any base open set

$N_a \subseteq \text{St}(\mathbb{B})$ for some $a \in \mathbb{B}$, since D is dense in $\mathbb{B}$, there exist some $d \in D$ with $d \leq a$, by Stone's Representation theorem, we have $N_d \subseteq N_a$. Hence $\mathcal{O}_D \cap N_a \neq \emptyset$ i.e. is dense in $\text{St}(\mathbb{B})$.

By assumption, we have $\bigcap_{D \in \mathcal{D}} \mathcal{O}_D$ is non-empty, let $\mathcal{U} \in \bigcap_{D \in \mathcal{D}} \mathcal{O}_D$. Now we claim $\mathcal{U} \in \mathbb{B}^+$ is $\mathcal{D}$ -generic. For any $D \in \mathcal{D}$ we have $\mathcal{U} \in \mathcal{O}_D$ then there exists $b \in D$ such that $\mathcal{U} \in N_b$, i.e., $b \in \mathcal{U}$. ■

Definition 1.11. A Boolean algebra $\mathbb{B}$ is **complete** if for all $S \subseteq \mathbb{B}$, both $\sup S = \bigvee S$ and $\inf S = \bigwedge S$ exist.

Theorem 1.12 (Forcing Theorem). Let M be a transitive model of ZFC, $\mathbb{P} \in M$ be a forcing poset, $G \subseteq \mathbb{P}$ be $\mathbb{P}$ -generic over M , $\varphi(\bar{x})$ a formula, $\tau_1, \dots, \tau_n \in M^{\mathbb{P}}$, then

$$\exists p \in G (p \Vdash \varphi(\bar{\tau})) \Leftrightarrow M[G] \models \varphi(\tau_1^G, \dots, \tau_n^G).$$

Now we recast the general setting of forcing in the language of complete Boolean algebras and Boolean values.

Let $\mathbb{B}$ be a complete Boolean algebra. Define sets $V_\alpha^{\mathbb{B}}$ by recursion on $\alpha \in \text{Ord}$:

$$V_\alpha^{\mathbb{B}} = \left\{ \tau : \left(\tau \text{ is a function with } \text{dom}(\tau) \subseteq \bigcup_{\beta < \alpha} V_\beta^{\mathbb{B}} \right) \wedge (\text{ran}(\tau) \subseteq \mathbb{B}) \right\}$$

Define $V^{\mathbb{B}} = \bigcup_{\alpha \in \text{Ord}} V_\alpha^{\mathbb{B}}$, the elements of $V_\alpha^{\mathbb{B}}$ are "the $\mathbb{B}$ -names of rank $\leq \alpha$ ".

Define for each formula φ and $\mathbb{B}$ -names $\tau_1, \dots, \tau_n$, we associate an element $\|\varphi(\tau_1, \dots, \tau_n)\|$ which we call it "Boolean value", this proceeds by recursion on the complexity of formulas:

First for atomic case, we define recursively on $(\text{rk}(\tau_1), \text{rk}(\tau_2))$ for every pair (τ_1, τ_2) of $\mathbb{B}$ -names, the Boolean values: $\|\tau_1 \in \tau_2\|, \|\tau_1 = \tau_2\|, \|\tau_1 \subseteq \tau_2\|$.

Here "by recursion" we meant $(\alpha_1, \beta_1) \leq (\alpha_2, \beta_2)$ if and only if $((\max(\alpha_1, \beta_1) < \max(\alpha_2, \beta_2)) \vee ((\max(\alpha_1, \beta_1) = \max(\alpha_2, \beta_2) \wedge (\alpha_1, \beta_1) \leq_{\text{lex}} (\alpha_2, \beta_2)))$.

$$\|\tau_1 \in \tau_2\| = \bigvee_{\sigma \in \text{dom}(\tau_2)} (\|\tau_1 = \sigma\| \wedge \tau_2(\sigma)), \|\tau_1 \subseteq \tau_2\| = \bigwedge_{\sigma \in \text{dom}(\tau_1)} (\tau_1(\sigma) \rightarrow \|\sigma \in \text{dom}(\tau_2)\|)$$

and $\|\tau_1 = \tau_2\| = \|\tau_1 \subseteq \tau_2\| \wedge \|\tau_2 \subseteq \tau_1\|$.

Then we extend this to all $\varphi(\bar{\nu})$:

$$\|\neg \varphi(\bar{\tau})\| = \neg \|\varphi(\bar{\tau})\|, \|\varphi \wedge \psi(\bar{\tau})\| = \|\varphi(\bar{\tau})\| \wedge \|\psi(\bar{\tau})\|, \|\exists x \psi(x, \bar{\tau})\| = \bigvee_{\sigma \in V^{\mathbb{B}}} \|\psi(\sigma, \bar{\tau})\|$$

Definition 1.13 (Forcing Relation). Let $b \in \mathbb{B}^+$, $\varphi(\bar{\nu})$ be a formula and $\tau_1, \dots, \tau_n \in V^{\mathbb{B}}$.

We define $b \Vdash^{\mathbb{B}} \varphi(\bar{\tau})$ if and only if $b \leq \|\varphi(\bar{\tau})\|$.

Note that $\|\varphi(\bar{\tau})\| = \bigvee \{b \in \mathbb{B}^+ : b \Vdash^{\mathbb{B}} \varphi(\bar{\tau})\}$.

We defined before $V^{\mathbb{P}}$ and $\Vdash^{\mathbb{P}}$ for every forcing $\mathbb{P}$, in case $\mathbb{P} = \mathbb{B}^+$, for a complete Boolean algebra $\mathbb{B}$, both approach are inter-translatable, and for every $\tau \in V^{\mathbb{P}}$, there exists $\tau' \in V^{\mathbb{B}^+}$ such that $\mathbb{1}_{\mathbb{P}} \Vdash \tau = \tau'$.

Theorem 1.14. Let M be a transitive model of ZFC, $\mathbb{B} \in M$ be a complete Boolean algebra, $G \subseteq \mathbb{B}^+$ be a generic filter over M , $\varphi(\bar{\nu})$ be a formula and $\tau_0, \dots, \tau_n \in M^{\mathbb{B}}$, then $\|\varphi(\bar{\tau})\| \in G$ if and only if $M[G] \models \varphi(\tau_1^G, \dots, \tau_n^G)$.

Before we proof this theorem, let us introduce some useful lemma.

Lemma 1.15. *Let M be a transitive model of ZFC, $\mathbb{B} \in M$ be a complete Boolean algebra. For a filter $\mathcal{F} \subseteq \mathbb{B}^+$, the followings are equivalent:*

1. $\mathcal{F} \subseteq \mathbb{B}^+$ is generic over M .
2. $(\forall S \subseteq \mathbb{B}^+ \text{ with } S \in M)(S \subseteq \mathcal{F} \text{ if and only if } \bigwedge S \in \mathcal{F} \text{ and } \mathcal{F} \text{ is an ultrafilter})$.
3. $(\forall S \subseteq \mathbb{B}^+ \text{ with } S \in M)(\mathcal{F} \cap S \neq \emptyset \text{ if and only if } \bigvee S \in \mathcal{F})$.

Proof. See Exercise Sheet 2. ■

Lemma 1.16. *For all $\tau_1, \tau_2 \in M^{\mathbb{B}}$,*

$$\|\tau_1 \in \tau_2\| \in G \Leftrightarrow \tau_1^G \in \tau_2^G; \|\tau_1 \subseteq \tau_2\| \in G \Leftrightarrow \tau_1^G \subseteq \tau_2^G; \|\tau_1 = \tau_2\| \in G \Leftrightarrow \tau_1^G = \tau_2^G$$

Proof. By recursion on $(\text{rk}(\tau_1), \text{rk}(\tau_2))$:

$$\begin{aligned} \|\tau_1 \in \tau_2\| \in G &\Leftrightarrow \bigvee_{\sigma \in \text{dom}(\tau_2)} (\|\sigma = \tau_1\| \wedge \tau_2(\sigma)) \in G \Leftrightarrow (\exists \sigma \in \text{dom}(\tau_2)) ((\sigma^G = \tau_1^G) \wedge (\tau_2(\sigma) \in G)) \Leftrightarrow \tau_1^G \in \tau_2^G \\ \|\tau_1 \subseteq \tau_2\| \in G &\Leftrightarrow \left(\bigwedge_{\sigma \in \text{dom}(\tau_1)} (\tau_1(\sigma) \rightarrow \|\sigma \in \text{dom}(\tau_2)\|) \right) \in G \Leftrightarrow (\forall \sigma \in \text{dom}(\tau_1)) ((\tau_1(\sigma) \rightarrow \|\sigma \in \tau_2\|) \in G) \Leftrightarrow \\ &(\forall \sigma \in \text{dom}(\tau_1)) (\tau_1(\sigma) \in G \rightarrow (\sigma^G \in \tau_2^G)) \Leftrightarrow \tau_1^G \subseteq \tau_2^G \end{aligned}$$
■

proof of the theorem. By induction on the complexity of φ .

Atomic case: follows from the previous lemma:

$$\begin{aligned} (\varphi \equiv \neg \psi) \quad \|\varphi(\bar{\tau})\| \in G &\Leftrightarrow \|(\neg \psi)(\bar{\tau})\| \in G \Leftrightarrow (\neg \|\psi(\bar{\tau})\|) \in G \\ &\Leftrightarrow \|\psi(\bar{\tau})\| \notin G \xleftrightarrow{\text{IH}} M[G] \not\models \psi(\bar{\tau}^G) \Leftrightarrow \neg(M[G] \models \psi(\bar{\tau}^G)) \\ &\Leftrightarrow M[G] \models \neg \psi(\bar{\tau}^G) \Leftrightarrow M[G] \models \varphi(\bar{\tau}^G) \\ (\varphi \equiv \psi \wedge \xi) \quad \|\varphi(\bar{\tau})\| \in G &\Leftrightarrow \|(\psi \wedge \xi)(\bar{\tau})\| \in G \Leftrightarrow \|\psi(\bar{\tau}) \wedge \xi(\bar{\tau})\| \in G \\ &\Leftrightarrow (\|\psi(\bar{\tau})\| \wedge \|\xi(\bar{\tau})\|) \in G \Leftrightarrow (\|\psi(\bar{\tau})\| \in G) \wedge (\|\xi(\bar{\tau})\| \in G) \\ &\xleftrightarrow{\text{IH}} (M[G] \models \psi(\bar{\tau}^G)) \wedge (M[G] \models \xi(\bar{\tau}^G)) \Leftrightarrow M[G] \models \varphi(\bar{\tau}^G). \\ (\varphi \equiv \exists x \psi(x, \bar{\nu})) \quad \|\varphi(\bar{\tau})\| \in G &\Leftrightarrow \|\exists x \psi(x, \bar{\tau})\| \in G \Leftrightarrow \left(\bigvee_{\sigma \in M^{\mathbb{B}}} \|\psi(\sigma, \bar{\tau})\| \right) \in G \\ &\Leftrightarrow (\exists \sigma \in M^{\mathbb{B}})(\|\psi(\sigma, \bar{\tau})\| \in G) \xleftrightarrow{\text{IH}} (\exists \sigma \in M^{\mathbb{B}})(M[G] \models \psi(\sigma^G, \bar{\tau}^G)) \\ &\Leftrightarrow M[G] \models \exists x \psi(x, \bar{\tau}^G) \Leftrightarrow M[G] \models \varphi(\bar{\tau}^G) \end{aligned}$$
■

Thus we have shown the Forcing Theorem (in the language of Boolean value), and the proof is much easier than the original version.

Definition 1.17. For $\mathbb{P}, \mathbb{Q}$ two forcing posets, a map $i: \mathbb{P} \rightarrow \mathbb{Q}$ is called **dense embedding** if

1. $i(1_{\mathbb{P}}) = 1_{\mathbb{Q}}$
2. $\forall p_1, p_2 \in \mathbb{P} ((p_1 \leq_{\mathbb{P}} p_2) \rightarrow (i(p_1) \leq_{\mathbb{Q}} i(p_2)))$,
3. $\forall p_1, p_2 \in \mathbb{P} ((p_1 \perp p_2) \rightarrow (i(p_1) \perp i(p_2)))$,
4. $\text{ran}(i)$ is dense in $\mathbb{Q}$.

Note that $i(p_1) \perp i(p_2)$ implies $p_1 \perp p_2$ (follows from 2).

Theorem 1.18. For every forcing poset $\mathbb{P}$, there exists $(\mathbb{B}, i)$ (unique upto isomorphic) such that $\mathbb{B}$ is a complete Boolean algebra and $i: \mathbb{P} \rightarrow \mathbb{B}^+$ is a dense embedding. $\mathbb{B}$ is called the Boolean completion of $\mathbb{P}$.

Proof idea 1. Define $\mathbb{B} = \{ \bigvee A : A \subseteq \mathbb{P} \}$. Inspired by the following proposition.

Proposition 1.19. $\mathbb{P} \stackrel{\text{dense}}{\subseteq} \mathbb{B}^+$ complete Boolean algebra then for $A, B \subseteq \mathbb{P}, \bigvee A \leq \bigvee B$ if and only if $\forall a \in A (\forall c \in \mathbb{P} \text{ with } c \leq a) (\exists b \in B) (c \parallel b)$, we say $A \leq_{\text{pre}} B \Leftrightarrow \bigvee A \leq \bigvee B$.

Proof. $\Rightarrow$: $\forall a \in A$, we have $a \leq \bigvee B$ thus $\forall c \in \mathbb{P}$ with $c \leq a, c \leq \bigvee B$. Then $c = c \wedge \bigvee B = \bigvee_{b \in B} c \wedge b$, since $c \neq 0$ thus $\exists b \in B$ such that $c \wedge b \neq 0$ which means $c \parallel b$. $\Leftarrow$: Suppose there exist some $a \in A$ such that $a \not\leq \bigvee B$ then $a \wedge \neg(\bigvee B) \neq 0$. Since $\mathbb{P}$ is dense so there exists $p \in \mathbb{P}$ with $p \leq a \wedge \neg(\bigvee B)$ thus $p \wedge \neg(\bigvee B) \neq 0$ but then contradict to our assumption that $p \parallel b$ for all $b \in B$. ■

Now we define an equivalence relation $\sim$ on $\mathcal{P}(\mathbb{P})$, $A \sim B \Leftrightarrow (A \leq_{\text{pre}} B) \wedge (B \leq_{\text{pre}} A)$. Then define $\bigvee A := [A]_{\sim} \in \mathcal{P}(\mathbb{P}) / \sim$ and $\bigvee A \leq \bigvee B$ if and only if $A \leq_{\text{pre}} B$.

Then $\mathbb{B}$ is a Boolean completion of $\mathbb{P}$. ■

Proof idea 2. Using algebra of regular open sets. Let (X, τ) be a topological space. An open set $A \subseteq X$ is regular if $\text{int}(\overline{A}) = A$. Note that $\text{int}(\overline{\text{int}(\overline{A})}) = \text{int}(\overline{A})$.

For $A \subseteq X$, define $\text{Reg}(A) = \text{int}(\overline{A})$, $\text{ro}(X, \tau) := \{ A \subseteq X : A = \text{Reg}(A) \}$, $A \wedge B = A \cap B$, $A \vee B = \text{Reg}(A \cup B)$, $\neg A = \text{int}(X \setminus A) = X \setminus \overline{A}$, $\bigvee_{i \in I} A_i = \text{Reg}(\bigcup_{i \in I} A_i)$, $\bigwedge_{i \in I} A_i = \text{Reg}(\bigcap_{i \in I} A_i)$. The order topology $\tau_{\leq}$ on $\mathbb{P}$ is the topology generated by the sets $p \downarrow$ for $p \in \mathbb{P}$ with $p \downarrow = \{ q \in \mathbb{P} : q \leq p \}$. ■

Lemma 1.20. $i: \mathbb{P} \rightarrow \text{ro}(\mathbb{P}, \tau_{\leq}) = \mathbb{B}, p \mapsto \text{int}(\overline{p \downarrow}) = \text{Reg}(p \downarrow)$ is a dense embedding.

Proof. • $i(1_{\mathbb{P}}) = \mathbb{B} = 1_{\mathbb{B}}$.

- $q \leq p \Rightarrow q \downarrow \subseteq p \downarrow \Rightarrow \text{Reg}(q \downarrow) \subseteq \text{Reg}(p \downarrow)$.
- $p \perp q \Leftrightarrow (p \downarrow) \cap (q \downarrow) = \emptyset \stackrel{*}{\Leftrightarrow} \text{Reg}(p \downarrow) \cap \text{Reg}(q \downarrow) = \emptyset \stackrel{**}{\Leftrightarrow} \text{Reg}(p \downarrow) \perp \text{Reg}(q \downarrow)$.
 $*$: $\Leftarrow$: as $q \downarrow \subseteq \text{Reg}(q \downarrow)$; as for $\Rightarrow$: $(p \downarrow) \cap (q \downarrow) = \emptyset \Rightarrow \overline{p \downarrow} \cap (q \downarrow) = \emptyset$ and $**$ follows from the same argument.

- Let $A(\neq \emptyset) \subseteq \mathbb{P}$ be regular open. There is $p \in \mathbb{P}$ such that $p \downarrow \subseteq A$, thus $\text{Reg}(p \downarrow) \subseteq \text{Reg}(A) = A$. Hence $\text{ran}(i)$ is dense in $\text{ro}(\mathbb{P}, \tau_{\leq}) = \mathbb{B}^+$. ■

Proof of the uniqueness of theorem 1.18. Suppose there exists $\mathbb{C}$ and a dense embedding $j: \mathbb{P} \rightarrow \mathbb{C}$. We want to show there exists two Boolean algebra isomorphism $g: \mathbb{B} \rightarrow \mathbb{C}, h: \mathbb{C} \rightarrow \mathbb{B}$ and they are hte inverse of each other.

For all $b \in \mathbb{B}$, define $g(b) = \left(\bigvee_{p \in \mathbb{P}, i(p) \leq b} j(p) \right)^{\mathbb{C}}$ by density of i and similarly define $h(c) = \left(\bigvee_{p \in \mathbb{P}, j(p) \leq c} i(p) \right)^{\mathbb{B}}$ for all $c \in \mathbb{C}$.

Then $\forall p \in \mathbb{P}, \forall b \in \mathbb{B}$, we have $i(p) \leq b \rightarrow j(p) \leq g(b)$ and $i(p) \perp b \rightarrow j(p) \perp g(b)$. Hence also $j(p) \leq g(b) \rightarrow i(p) \leq b$, if not, then there exists some $r \in \mathbb{P}$ with $i(r) \leq i(p)$ and $i(r) \perp b$, by the property of dense embedding we have $r \parallel p$ and thus $j(r) \parallel j(p), j(r) \perp g(b)$, but then contradict to $j(p) \leq g(b)$.

Hence $\forall b \in \mathbb{B}, b = \sup\{i(p) : p \in \mathbb{P} \wedge i(p) \leq b\} = \sup\{i(p) : p \in \mathbb{P} \wedge j(p) \leq g(b)\} = h(g(b))$. ■

So given $\mathbb{P}, i: \mathbb{P} \rightarrow \mathbb{B}^+$ here $\mathbb{B}$ is a complete Boolean algebra, $p \Vdash^{\mathbb{P}} \varphi(\bar{\tau}) \iff i(p) \Vdash^{\mathbb{B}} \varphi(\bar{\sigma})$ for some $\sigma \in V^{\mathbb{B}}$ corresponding to $\bar{\tau}$.

Proposition 1.21.

$$\forall \tau_1, \tau_2, \tau_3 \in V^{\mathbb{B}}, \|\tau_1 = \tau_1\| = \mathbb{1}_{\mathbb{B}}, \|\tau_1 = \tau_2\| = \|\tau_2 = \tau_1\|, \|\tau_1 = \tau_2\| \wedge \|\tau_2 = \tau_3\| \leq \|\tau_1 = \tau_3\|,$$

$$\|\tau_1 = \tau_2\| \wedge \|\tau_2 \in \tau_3\| \leq \|\tau_1 \in \tau_3\|, \|\tau_1 \in \tau_2\| \wedge \|\tau_2 = \tau_3\| \leq \|\tau_1 \in \tau_3\|.$$

Proof. We show $\|\tau = \tau\| = 1$ by induction on $\text{rk}(\tau)$. $\|\tau \subseteq \tau\| = \bigwedge_{\sigma \in \text{dom}(\tau)} (\tau(\sigma) \rightarrow \|\sigma \in \tau\|)$.

Let $\sigma \in \text{dom}(\tau)$ then $\tau(\sigma) \rightarrow \|\sigma \in \tau\| \equiv \neg \tau(\sigma) \vee \|\sigma \in \tau\| = \neg \tau(\sigma) \vee \bigvee_{\pi \in \text{dom}(\tau)} (\tau(\pi) \wedge \|\sigma = \pi\|)$, by induction hypothesis, $\bigvee_{\pi \in \text{dom}(\tau)} (\tau(\pi) \wedge \|\sigma = \pi\|) \geq \pi(\sigma)$ thus $\tau(\sigma) \rightarrow \|\sigma \in \tau\| \geq \neg \tau(\sigma) \vee \tau(\sigma) = 1$. ■

Lemma 1.22. Let $\mathbb{P}$ be a forcing poset and $A \subseteq \mathbb{P}$ be an antichain. For each $q \in A$, let σ_q be a $\mathbb{P}$ -name. There then exists a $\mathbb{P}$ -name τ such that $\forall q \in A, q \Vdash \tau = \sigma_q$.

Proof. Define $\tau = \bigcup_{q \in A} \bigcup_{(\pi, r) \in \sigma_q} \{(\pi, r') : (r' \in \mathbb{P}) \wedge (r' \leq q, r)\}$. Let $q \in A$ and $G \subseteq \mathbb{P}$ be $\mathbb{P}$ -generic over V with $q \in G$. We claim $\tau^G = \sigma_q^G$.

For $\subseteq$: If $a \in \tau^G$ then $(\exists \hat{q} \in A)(\exists (\pi, r) \in \sigma_{\hat{q}})(\exists r' \leq \hat{q}, r)$ with $r' \in G$ and $a = \pi^G$. So let $\hat{q}, (\pi, r)$ and r' be such. Then $\hat{q} \in G, r \in G, r \leq q$ thus $q \in G$, since $q, \hat{q}$ both in the antichain so $q = \hat{q}$. Hence $a = \pi^G \in \sigma_q^G$.

For $\supseteq$: If $a \in \sigma_q^G$ then there exists $(\pi, r) \in \sigma_q$ with $r \in G$, since q, r both in the generic filter G so there exists $r' \in G$ with $r' \leq r, q$ so $a \in \tau^G$ by definition. ■

Theorem 1.23. Let $\varphi(x_0, \dots, x_n, x)$ be a formula, $\tau_0, \dots, \tau_n$ be $\mathbb{P}$ -names, $p \in \mathbb{P}$, then $p \Vdash \exists x \varphi(\tau_0, \dots, \tau_n, x)$ if and only if there exists $\sigma \in V^{\mathbb{P}}$ such that $p \Vdash \varphi(\tau_0, \dots, \tau_n, \sigma)$.

Proof. For $\Leftarrow$ it is clear. Now let us show $\Rightarrow$, let $\mathcal{G}$ consists of all $A \subseteq p \downarrow$ such that A is an antichain and $\forall q \in A, \exists \sigma \in V^{\mathbb{P}}, q \Vdash \varphi(\bar{\tau}, \sigma)$. We claim $\mathcal{G} \neq \emptyset$: let $q \leq p$ and $G \subseteq \mathbb{P}$ be a $\mathbb{P}$ -generic filter with

$q \in G$. Then $V[G] \models \exists x \varphi(\bar{\tau}^G, x)$. So for some $\sigma \in V^{\mathbb{P}}$, $V[G] \models \varphi(\bar{\tau}^G, \sigma^G)$, by forcing theorem there exists some $r \in G$ such that $r \Vdash \varphi(\bar{\tau}, \sigma)$, then we can find some $r' \leq r, q$ with $r' \Vdash \varphi(\bar{\tau}, \sigma)$. **But this only explained why there exists some condition q below p and some $\mathbb{P}$ -name σ such that q forces $\varphi(\bar{\tau}, \sigma)$, but why there exist an antichain which consist of such conditions?**

Apply Zorn's lemma on $\mathcal{G}$, we obtained an maximal element A of $(\mathcal{G}, \subseteq)$. For each $q \in A$, fix some $\sigma_q \in V^{\mathbb{P}}$ such that $q \Vdash \varphi(\bar{\tau}, \sigma_q)$. Apply the previous lemma, choose some $\sigma \in V^{\mathbb{P}}$ such that $q \Vdash \sigma_q = \sigma$ for all $a \in A$.

Now we claim $p \Vdash \varphi(\bar{\tau}, \sigma)$. If not, then find $r \in p \downarrow$ such that $r \Vdash \neg \varphi(\bar{\tau}, \sigma)$. Then $r \perp q$ for all $q \in A$, find $r' \leq r, \sigma_{r'} \in V^{\mathbb{P}}$ such that $r' \Vdash \varphi(\bar{\tau}, \sigma_{r'})$. Then contradict to the maximality of A . $\blacksquare$

Definition 1.24. $D \subseteq \mathbb{P}$ is called **predense** if $\forall p \in \mathbb{P}, \exists d \in D, d \parallel p$.

Definition 1.25. Let $\mathbb{P}, \mathbb{Q}$ be forcing posets, $i: \mathbb{P} \rightarrow \mathbb{Q}$ is a **complete embedding** if

1. $i(1_{\mathbb{P}}) = 1_{\mathbb{Q}}$,
2. $(*)$ If $A \subseteq \mathbb{P}$ is an maximal antichain then so is $i''A \subseteq \mathbb{Q}$,
3. $(\forall p, q \in \mathbb{P})(((p \leq_{\mathbb{P}} q) \rightarrow (i(p) \leq_{\mathbb{Q}} i(q))) \wedge ((p \perp q) \rightarrow (i(p) \perp i(q))))$ holds.

And we can replace 2 by $(**)$ If $D \subseteq \mathbb{P}$ is predense then so is $i''D \subseteq \mathbb{Q}$.

Now let us check the equivalence: For $(*) \rightarrow (**)$, let $D \subseteq \mathbb{P}$ be predense, now suppose there exists $q \in \mathbb{Q}$ such that $\forall d \in D, i(d) \perp q$. Set $D \downarrow := \{p \in \mathbb{P}: (\exists d \in D)(p \leq d)\}$, since D is predense so $D \downarrow$ is open dense. Let $A \subseteq D \downarrow$ be an maximal antichain, actually A is also maximal in $\mathbb{P}$ because suppose there exists $p \in \mathbb{P}$ with $p \perp a$ for all $a \in A$, but then there exists $d \in D \downarrow$ with $d \leq p$ and $d \perp a$ for all $a \in A$ which contradict to the maximality of A . Since i preserves maximal antichain, $\forall a \in A, \exists d_a \in D$ such that $a \leq_{\mathbb{P}} d_a$. Now we have $i(a) \leq_{\mathbb{Q}} i(d_a)$ and $i(d_a) \perp q$, thus $i(a) \perp q$ then contradict to the maximality of $i''A$ which leads to a contradiction.

For $(**) \rightarrow (*)$, let $A \subseteq \mathbb{P}$ be an maximal antichain, we want to show $i''A \subseteq \mathbb{Q}$ is still maximal. Suppose not, there exists $q \in \mathbb{Q}$ such that $q \perp i(a)$ for all $a \in A$. Set $A \downarrow := \{p \in \mathbb{P}: \exists a \in A, p \leq a\}$ and it is dense in $\mathbb{P}$ hence also predense. Since $i''(A \downarrow) \subseteq \mathbb{Q}$ is predense there exists $d \in A \downarrow$ such that $i(d) \parallel q$. And by construction of $A \downarrow$, there exists $a_d \in A$ such that $d \leq_{\mathbb{P}} a_d$, so $i(d) \leq i(a_d)$. But $q \perp i(a_d)$ thus $q \perp i(d)$ which leads to a contradiction.

Lemma 1.26. Let $i: \mathbb{P} \rightarrow \mathbb{Q}$ be a complete embedding and $G \subseteq \mathbb{Q}$ be a $\mathbb{Q}$ -generic filter over V then $i^{-1}G \subseteq \mathbb{P}$ is a $\mathbb{P}$ -generic filter over V .

Proof. Set $H := i^{-1}G$. Let $A \subseteq \mathbb{P}, D \in V$ be an maximal antichain, then $i''A$ is also an maximal antichain in $\mathbb{Q}$, thus $G \cap i''A \neq \emptyset$. Say $q \in i''A \cap G, q = i(r)$ for some $r \in A$, then $r \in A \cap H$. Let $p, p' \in H$ then $i(p), i(p') \in G$. Denote $p^{\perp} := \{q \in \mathbb{P}: q \perp p\}$. Then $D = p^{\perp} \cup p'^{\perp} \cup (p \downarrow \cap p' \downarrow)$ is dense. So $H \cap D \neq \emptyset$, say $q \in D \cap H$. If $q \in p^{\perp}$ (or $p'^{\perp}$ similar argument holds) then $q \perp p$ and $i(q) \perp i(p)$, but $i(q), i(p) \in G$ which contradict to G is a filter thus $q \in p \downarrow \cap p' \downarrow \cap H$. $\blacksquare$

Note: If $i: \mathbb{P} \rightarrow \mathbb{Q}$ is a complete(dense) embedding and $G \subseteq \mathbb{Q}$ is $\mathbb{Q}$ -generic filter over V . Define $H = i^{-1''}G$ then $V[H] \subseteq V[G]$ ($V[H] = V[G]$). Because H is defined by using i and G , and i is a function in V so H must be in $V[G]$.

Definition 1.27. Define recursively for any $i_*: V^{\mathbb{P}} \rightarrow V^{\mathbb{Q}}$ by $i_*(\tau) = \{(i_*(\sigma), i(p)) : (\sigma, p) \in \tau\}$. By induction, every $\tau \in V^{\mathbb{P}}, i_*(\tau) \in V^{\mathbb{Q}}$.

Lemma 1.28. If $i: \mathbb{P} \rightarrow \mathbb{Q}$ is a complete embedding, $G \subseteq \mathbb{Q}$ is a $\mathbb{Q}$ -generic filter over V . Define $H = i^{-1'}G$ then $\forall \tau \in V^{\mathbb{P}}, \tau^H = i_*(\tau)^G$.

Note that every dense embedding is also complete.

Let us check this: let $f: \mathbb{P} \rightarrow \mathbb{Q}$ be a dense embedding, we want to show f is complete, i.e., to show if $D \subseteq \mathbb{Q}$ is predense then $f''D$ is also predense in $\mathbb{Q}$. Let $q \in \mathbb{Q}$, we want to find some $d \in D$ such that $f(d) \parallel q$. Since $\text{ran}(f)$ is dense in $\mathbb{Q}$ so there exists $p \in \mathbb{P}$ such that $f(p) \leq_{\mathbb{Q}} q$ also D is predense so there exists $d \in D$ with $d \parallel p$. So $f(d) \parallel f(p)$ and hence $f(d) \parallel q$.

Lemma 1.29. Let $i: \mathbb{P} \rightarrow \mathbb{Q}$ be a dense embedding.

1. If $H \subseteq \mathbb{P}$ is a $\mathbb{P}$ -generic filter over V then $i''H \uparrow$ is $\mathbb{Q}$ -generic over V . For $G := i''H$, we have $i^{-1''}G = H$ and $V[H] = V[G]$.
2. If $G \subseteq \mathbb{Q}$ is a $\mathbb{Q}$ -generic filter over V then $i^{-1''}G$ is a $\mathbb{P}$ -generic over V . For $H := i^{-1''}G$, we have $i''H \uparrow = G$ and $V[H] = V[G]$.
3. For $p \in \mathbb{P}, \varphi$ a formula, $\tau_0, \dots, \tau_{n-1} \in V^{\mathbb{P}}, p \Vdash^{\mathbb{P}} \varphi(\tau_0, \dots, \tau_{n-1}) \iff i(p) \Vdash^{\mathbb{Q}} \varphi(i_*(\tau_0), \dots, i_*(\tau_{n-1}))$.

Proof. 1. Let $G = i''H \uparrow$, one can directly check that G is $\mathbb{Q}$ -generic filter (See Sheet 1 Exercise 2).

Now we show that $H = i^{-1''}G$. Since for any $p \in \mathbb{P} \setminus H$, there exists $h \in H, h \perp p$. This is because otherwise consider the dense set $D = \{q \in \mathbb{P} : q \leq p \vee q \perp p\}, D \cap H \neq \emptyset$, say $h' \in D \cap H$ then $h' \leq p$ which leads to a contradiction. If $H' \subseteq \mathbb{P}$ is a $\mathbb{P}$ -generic filter over V and $H' \neq H$ then there exists $p \in H$ and $p' \in H'$ such that $p \perp p'$. So if $i^{-1''}G \neq H$ then there exists $p' \in i^{-1''}G, p \in H$ such that $p' \perp p$ but then $i(p') \perp i(p)$ and this contradicts the fact $i(p') \parallel i(p)$.

2. Similar.

3. We prove $\Rightarrow$. Assume $p \Vdash^{\mathbb{P}} \varphi(\bar{\tau})$, we want to show $i(p) \Vdash^{\mathbb{Q}} \varphi(i_*(\bar{\tau}))$. Let $G \subseteq \mathbb{Q}$ be a $\mathbb{Q}$ -generic filter over V with $i(p) \in G$. Define $H := i^{-1''}G$ and then H is $\mathbb{P}$ -generic over V with $p \in H$, so $V[H] \models \varphi(\tau_0^H, \dots, \tau_{n-1}^H)$. Hence $V[G] \models \varphi(i_*(\tau_0)^G, \dots, i_*(\tau_{n-1})^G)$.

$\Leftarrow$ is similar.

■

Remark 1.30. It follows that if $\mathbb{P}, \mathbb{Q}$ are forcing posets and there exists a third forcing poset $\mathbb{R}$ such that $\mathbb{P}, \mathbb{Q}$ both embed densely into $\mathbb{R}$, then forcing with $\mathbb{P}$ is equivalent to forcing with $\mathbb{Q}$.

If $i: \mathbb{P} \rightarrow \mathbb{R}$ is a complete embedding and H is $\mathbb{P}$ -generic over V , K is $\mathbb{R}$ -generic over V then $V \subseteq V[H] \subseteq V[K]$. In fact $V[K]$ is a forcing extension of $V[H]$, i.e., there exists $\mathbb{Q} \in V[H]$ and $G \subseteq \mathbb{Q}$ a $\mathbb{Q}$ -generic filter over V such that $V[K] = V[H][G]$. Then there exists $\dot{\mathbb{Q}} \in V^{\mathbb{P}}$ such that $\mathbb{Q} = \dot{\mathbb{Q}}^H$.

Suppose $\mathbb{P}$ is a forcing poset and $\dot{\mathbb{Q}}$ is a $\mathbb{P}$ -name such that $\mathbb{1}_{\mathbb{P}} \Vdash^{\mathbb{P}} \dot{\mathbb{Q}}$ is a forcing poset then $V \subseteq V[G] \subseteq V[G][H]$. In fact $V[G][H]$ is a forcing extension of V , which means there is a forcing poset $\mathbb{R} \in V$ and K a $\mathbb{R}$ -generic filter over V such that $V[K] = V[G][H]$.

Definition 1.31. Let $\mathbb{P}$ be a forcing poset, $\dot{\mathbb{Q}}$ a $\mathbb{P}$ -name for a forcing poset. Then $\mathbb{P} * \dot{\mathbb{Q}}$ is the following forcing poset: $(\mathbb{R}, \leq, \mathbb{1}_{\mathbb{R}})$, where

$$\mathbb{R} = \{(p, \dot{q}) : p \in \mathbb{P}, \dot{q} \in \text{dom}(\dot{\mathbb{Q}}), p \Vdash \dot{q} \in \dot{\mathbb{Q}}\}, \mathbb{1}_{\mathbb{R}} = (\mathbb{1}_{\mathbb{P}}, \mathbb{1}_{\dot{\mathbb{Q}}})$$

and $(p_1, \dot{q}_1) \leq (p_2, \dot{q}_2)$ if and only if $(p_1 \leq_{\mathbb{P}} p_2) \wedge (\dot{q}_1 \leq_{\dot{\mathbb{Q}}} \dot{q}_2)$, here actually the rightmost $\leq$ is not an order yet, but we can ask what forces this, when we say $\dot{q}_1 \leq_{\dot{\mathbb{Q}}} \dot{q}_2$ what we meant was $p_1 \Vdash^{\mathbb{P}} \dot{q}_1 \leq_{\dot{\mathbb{Q}}} \dot{q}_2$.

We will confuse $\mathbb{P}$ -name $\dot{\mathbb{Q}}$ such that $\mathbb{1}_{\mathbb{P}} \Vdash^{\mathbb{P}} \dot{\mathbb{Q}}$ is a forcing poset with the triple $(\dot{\mathbb{Q}}, \leq_{\dot{\mathbb{Q}}}, \mathbb{1}_{\dot{\mathbb{Q}}})$ such that

- $\mathbb{1}_{\dot{\mathbb{Q}}} \in \text{dom}(\dot{\mathbb{Q}})$
- $\mathbb{1}_{\mathbb{P}} \Vdash^{\mathbb{P}} \leq_{\dot{\mathbb{Q}}}$ is a preorder on $\dot{\mathbb{Q}}$
- $\mathbb{1}_{\dot{\mathbb{Q}}} \in \dot{\mathbb{Q}}$ is a largest element of $(\dot{\mathbb{Q}}, \leq_{\dot{\mathbb{Q}}})$

Lemma 1.32. The map $i: \mathbb{P} \rightarrow \mathbb{P} * \dot{\mathbb{Q}}$ is a complete embedding.

Proof. • $i(\mathbb{1}_{\mathbb{P}}) = (\mathbb{1}_{\mathbb{P}}, \mathbb{1}_{\dot{\mathbb{Q}}})$.

- $\forall p_1, p_2 \in \mathbb{P}, p_2 \leq p_1 \Leftrightarrow i(p_2) \leq i(p_1)$ and $p_2 \perp p_1$ implies $i(p_2) \perp i(p_1) = (p_2, \mathbb{1}_{\dot{\mathbb{Q}}}) \perp (p_1, \mathbb{1}_{\dot{\mathbb{Q}}})$.
- Let $D \subseteq \mathbb{P}$ be predense and $(p, \dot{q}) \in \mathbb{P} * \dot{\mathbb{Q}}$. Let $p' \in D$ be such that $p' \parallel p$ then $i(p') \parallel (p, \dot{q})$ as there exists $p'' \leq p, p'$ and $p'' \Vdash \dot{q} \leq \mathbb{1}_{\dot{\mathbb{Q}}}, p'' \Vdash \dot{q} \leq \dot{q}$ thus $(p'', \dot{q}) \leq i(p'), (p, \dot{q})$.

■

Theorem 1.33. Let $\mathbb{P}$ be a forcing poset and $\dot{\mathbb{Q}}$ be a $\mathbb{P}$ -name for a forcing poset, then

1. Let K be a $\mathbb{P} * \dot{\mathbb{Q}}$ -generic filter over V and $G := i^{-1}K, H = \{\dot{q}^G : (p, \dot{q}) \in K\}$ then G is $\mathbb{P}$ -generic filter over V and H is $\dot{\mathbb{Q}}^G$ -generic filter over $V[G]$ and $V[K] = V[G][H]$.
2. Let G be a $\mathbb{P}$ -generic filter over V and H be a $\dot{\mathbb{Q}}^G$ -generic filter over $V[G]$, then $K = G * H = \{(p, \dot{q}) : p \in G \wedge \dot{q}^G \in H\}$ is $\mathbb{P} * \dot{\mathbb{Q}}$ -generic over V and $V[G][H] = V[K]$.

2 Applications

2.1 A model of ZFC with no "definable" well-order of reals and a model of ZF in which there is a countable set of countable sets of reals whose union is uncountable.

2.1.1 A model of ZFC with no "definable" well-order of reals

Definition 2.1. A set a is called **ordinal definable (OD)** if and only if there is a formula φ and ordinals $\alpha_0, \dots, \alpha_{n-1}$ such that

$$x \in a \iff \varphi(x, \alpha_0, \dots, \alpha_{n-1}).$$

Theorem 2.2 (Levy-Montague Reflection Principle). For every formula φ there is a club class of α such that for every such α and for all $x_0, \dots, x_{n-1} \in V_\alpha$,

$$\varphi(x_0, \dots, x_{n-1}) \iff V_\alpha \models \varphi(x_0, \dots, x_{n-1}).$$

Lemma 2.3. A set a is OD if and only if there is a formula φ and ordinals $\alpha_0, \dots, \alpha_{n-1} < \alpha$ such that $a \in V_\alpha$ and for all x ,

$$x \in a \iff V_\alpha \models \varphi(x, \alpha_0, \dots, \alpha_{n-1}).$$

Hence "Ordinal Definable" is definable.

Definition 2.4. A set a is called **hereditarily ordinal definable (HOD)**, $a \in \text{HOD}$ if and only if $\text{TC}(\{a\}) \subseteq \text{OD}$.

The class of sets which are hereditarily ordinal definable is called HOD, is definable. (**Check**)

Theorem 2.5 (Gödel). $\text{HOD} \models \text{ZFC}$.

Proof. Note that HOD is transitive, we then construct HOD as the $\mathcal{L}_\in$ -model $(\text{HOD}, \in \upharpoonright_{\text{HOD}})$.

Then (Ext), (Fund) holds trivially, since $\omega \in \text{HOD}$, (Inf) holds, (Pair) and (Union) are easy. For (Pow): let $a \in \text{HOD}$, we want to find $A \in \text{HOD}$ such that $\text{HOD} \models "A = \mathcal{P}(a)"$, here A may not be the set of all subset of a , instead, we want A to be the set of all ordinal definable subsets of A . Let $A = \{x \subseteq a : x \text{ is OD}\}$. As $a \in \text{HOD}$ and "OD" is definable, A is also OD, hence $A \in \text{HOD}$.

For (Aus), let $a \in \text{HOD}$, φ a formula and $b_0, \dots, b_{n-1} \in \text{HOD}$. Then $\bar{a} = \{x \in a : \text{HOD} \models \varphi(x, b_0, \dots, b_{n-1})\}$ is OD, hence $\bar{a} \in \text{HOD}$.

For (Rep), we will show axiom schema of collection which is equivalent to (Rep). Pick an ordinal $\alpha \in \text{HOD}$ big enough such that every $x \in a$ where $a \in \text{HOD}$ replaced by some φ by y , y is contained in $V_\alpha \cap \text{HOD}$. And clearly $V_\alpha \cap \text{HOD}$ is in HOD.

For (AC), let a be ordinal definable, say $x \in a$ if and only if $V_\alpha \models \varphi(x, \alpha_0, \dots, \alpha_{n-1})$ (take the least Gödel number of φ which does the job). Since there is a canonical well-order of finite sets of ordinals: let b, d be finite sets of ordinals, let $b <^* d$ if and only if $\max(b \triangle d) \in d$. $<^*$ is a well-order (Proof see Sheet 5 Exercise 1). In this fashion, each $a \in \text{HOD}$ has a canonical "name" where we minimize $\varphi, \alpha, \{\alpha_0, \dots, \alpha_{n-1}\}$. If a, b are both ordinal definable, say $a \tilde{<} b$ if and only if the canonical "name" of $a <$ the canonical "name" of b . Since $\tilde{<}$ is definable, hence $\tilde{<} \upharpoonright_a \in \text{HOD}$ for all $a \in \text{HOD}$. ■

Definition 2.6. Let c be a set. We say that a set a is ordinal definable from members of c if and only if there is a formula φ , ordinals $\alpha_0, \dots, \alpha_{n-1}$ and elements $p_0, \dots, p_{m-1} \in c$ such that for all x ,

$$x \in a \iff \varphi(x, \alpha_0, \dots, \alpha_{n-1}, p_0, \dots, p_{m-1}).$$

Definition 2.7. Let c be a set, we say a is hereditarily ordinal definable from elements of c if and only if every element of $\text{TC}(\{a\})$ is ordinal definable from elements of c , denoted by $a \in \text{HOD}_c$.

Theorem 2.8. Let c be a set such that c is ordinal definable from elements of c then $\text{HOD}_c \models \text{ZF}$.

Proof. Same as before except (Pow), let $a \in \text{HOD}_c$ and $A = \{x \subseteq a : x \in \text{OD}_c\}$. As $c \in \text{OD}_c$, so $A \in \text{OD}_c$. Hence $A \in \text{HOD}_c$. (This proof is extremely brief, actually we used Reflection Principle: there is some ordinal α such that for any $y \in \mathcal{P}(a) \cap \text{OD}_c \iff$ there is a formula φ , ordinal $\alpha_0, \dots, \alpha_{n-1}$ and $y_0, \dots, y_{m-1}$ elements from c such that $\forall u (u \in y) \leftrightarrow V_\alpha \models \varphi(u, \alpha_0, \dots, \alpha_{n-1}, y_0, \dots, y_{m-1})$. Since $c \in \text{OD}_c$ so $\mathcal{P}(a) \cap \text{OD}_c \in \text{OD}_c$. And it is easy to verify $A \in \text{HOD}_c$.) ■

Now let us construct counterexamples of Axiom of Choice in HOD_c .

Recall: $\mathbb{C}$ is the set of all finite sequence of natural numbers, ordered by end-extension, and the finite support product of α copies of $\mathbb{C}$, i.e., $\mathbb{C}(\alpha) = \{p \mid p : \alpha \rightarrow \overline{\mathbb{C}} \wedge \overline{\{i < \alpha \mid p(i) \neq 1_{\mathbb{C}}\}} < \aleph_0\}$. For $p \in {}^\alpha \mathbb{C}$, we write $\text{supp}(p) = \{i < \alpha : p(i) \neq 1_{\mathbb{C}}\}$, then $p \in \mathbb{C}(\alpha)$ if and only if $p \in {}^\alpha \mathbb{C}$ and $\overline{\text{supp}(p)} < \aleph_0$. From last semester we know both $\mathbb{C}$ and $\mathbb{C}(\alpha)$ have the c.c.c.

Definition 2.9. Let $\mathbb{P}$ be a poset, we call $\mathbb{P}$ **homogeneous** if for all $p, q \in \mathbb{P}$ there is an isomorphism (or dense endomorphism) π such that $\pi(p) \parallel q$.

Lemma 2.10. $\mathbb{C}$ is homogeneous.

Proof. Let $p, q \in \mathbb{C}$ be given, define

$$\pi(r) = \begin{cases} r, & \text{if none of the following is true: } r \subseteq p, p \subseteq r, r \subseteq q, q \subseteq r \\ q \restriction \text{dom}(r), & \text{if } r \subseteq p \\ p \restriction \text{dom}(r), & \text{if } r \subseteq q \\ q \cup (r \restriction [\text{dom}(q), \text{dom}(r)]), & \text{if } p \subseteq r \\ p \cup (r \restriction [\text{dom}(p), \text{dom}(r)]), & \text{if } q \subseteq r \end{cases}$$

Then π is what we want. ■

Lemma 2.11. $\mathbb{C}(\alpha)$ is homogeneous for all α .

Proof. Do the same thing on $\xi \in \text{supp}(p) \cap \text{supp}(q)$. ■

Lemma 2.12. Let M be a transitive model of ZFC and $\mathbb{P} \in M$ be a poset such that $M \models \text{"}\mathbb{P}\text{ is homogeneous"}$.

Let $\pi \in M, \pi : \mathbb{P} \rightarrow \mathbb{P}$ be an isomorphism, then π induces a map $\tilde{\pi} : M^{\mathbb{P}} \rightarrow M^{\mathbb{P}}$ (recursively) defined by $\tilde{\pi}(\tau) = \{(\tilde{\pi}(\sigma), \pi(p)) : (\sigma, p) \in \tau\}$ where $\tau \in M^{\mathbb{P}}$ with the following property: for every formula φ and for all $p \in \mathbb{P}$ and for all $\tau_1, \dots, \tau_n \in M^{\mathbb{P}}, p \Vdash \varphi(\tau_1, \dots, \tau_n) \iff \pi(p) \Vdash \varphi(\tilde{\pi}(\tau_1), \dots, \tilde{\pi}(\tau_n))$.

Proof. We may assume M to be countable.

Claim: let $g \subseteq \mathbb{P}$ be M -generic and $h := \pi''g \subseteq \mathbb{P}$, then h is also M -generic and $\tau^g = \tilde{\pi}(\tau)^h$. (If π is dense endomorphism then let $h := \pi''g \uparrow$.)

Proof of the claim: $x \in \tau^g$ if and only if there is some $p \in g$ and some σ with $(\sigma, p) \in \tau$. And $x \in \tilde{\pi}(\tau)^h$ if and only if there is some $\bar{p} \in h$ and some $\bar{\sigma}$ with $(\bar{\sigma}, \bar{p}) \in \tilde{\pi}(\tau)$. Take $\bar{p} = \pi(p)$, $\bar{\sigma} = \tilde{\pi}(\sigma)$.

And same argument shows the claim: let $g \subseteq \mathbb{P}$ be M -generic and $h = \pi^{-1''}g$ then h is also M -generic and $\tau^h = \tilde{\pi}(\tau)^g$.

Now we show $\Rightarrow$, let $g \subseteq \mathbb{P}$ be any M -generic with $\pi(p) \in g$, let $h = \pi^{-1''}g$ then $p \in h$. And thus $M[h] \models \varphi(\tau_1^h, \dots, \tau_n^h)$, since $M[g] = M[h]$, we have $M[g] \models \varphi(\tilde{\pi}(\tau_1)^g, \dots, \tilde{\pi}(\tau_n)^g)$ thus $\pi(p) \Vdash \varphi(\tilde{\pi}(\tau_1), \dots, \tilde{\pi}(\tau_n))$. Converse holds by the same argument. $\blacksquare$

Theorem 2.13. *Let M be a transitive model of ZFC and $g \subseteq \mathbb{C}$ be M -generic. Then $\bigcup g \in {}^\omega\omega$ is not definable over $M[g]$ from any element of M .*

Proof. Assume there is some formula φ and some $x_1, \dots, x_n \in M$ such that $(\bigcup g)(n) = m$ if and only if $M[g] \models \varphi(n, m, x_1, \dots, x_n)$ for all $n, m < \omega$. Let $\pi: \mathbb{C} \rightarrow \mathbb{C}$ be any isomorphism, $\pi \in M$. For all $x \in M$, $\tilde{x} = \tilde{\pi}(x)$. Hence if $p, q \in \mathbb{C}$ and $\pi \in M$ is an isomorphism with $\pi(p) \parallel q$ then

$$p \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k) \iff \pi(p) \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$$

If $p \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$, $q \Vdash \neg \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$ but $\pi(p) \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$ contradict to $\pi(p) \parallel q$. So there are no p, q with $p \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$, $q \Vdash \neg \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$, hence $\mathbb{1}_{\mathbb{C}} \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$ or $\mathbb{1}_{\mathbb{C}} \Vdash \neg \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$.

Therefore $(\bigcup g)(n) = m \iff \exists p \in g, p \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k) \iff \mathbb{1}_{\mathbb{C}} \Vdash \varphi(\tilde{n}, \tilde{m}, x_1, \dots, x_k)$, then $\bigcup g \in M$ which contradict to the fact that $\mathbb{C}$ is atomless. $\blacksquare$

A similar argument shows the following:

Theorem 2.14. *Let M be a transitive model of ZFC and $\alpha \in M, g \subseteq \mathbb{C}(\alpha)$ be a M -generic filter.*

Let $\chi_\xi = \bigcup \{p(\xi) : p \in g\} (= \bigcup g_\xi) \in {}^\omega\omega$ for $\xi < \alpha$, then $\chi_\xi \neq \chi_{\xi'}$ for any $\xi \neq \xi'$.

Proof. Suppose not, let $\xi \neq \xi'$ be such that $M[g] \models \chi_\xi = \chi_{\xi'}$. By forcing theorem, there exists $p \in g$ such that $p \Vdash \dot{\chi}_\xi = \dot{\chi}_{\xi'}$. But $\mathbb{C}(\alpha)$ is homogeneous, so for any $p, q \in \mathbb{C}(\alpha)$, there exists an isomorphism π such that $\pi(p) \parallel q$, then by the similar argument, $1 \Vdash \chi_\xi(\tilde{n}) = \chi_{\xi'}(\tilde{n})$ or $1 \Vdash \chi_\xi(\tilde{n}) \neq \chi_{\xi'}(\tilde{n})$. By assumption we have $1 \Vdash \chi_\xi(\tilde{n}) = \chi_{\xi'}(\tilde{n})$.

However, we can easily construct a condition $q \leq 1$ that forces them to be different. Define a condition $q \in \mathbb{C}(\alpha)$ such that: $q(\gamma) = \emptyset$ for $\gamma \neq \xi, \xi'$, $q(\xi) = \{(n, 0)\}$, $q(\xi') = \{(n, 1)\}$. Clearly $q \leq 1$. But $q \Vdash \dot{\chi}_\xi(\tilde{n}) = 0$ and $q \Vdash \dot{\chi}_{\xi'}(\tilde{n}) = 1$, so:

$$q \Vdash \dot{\chi}_\xi \neq \dot{\chi}_{\xi'}$$

This contradicts the fact that $1 \Vdash \dot{\chi}_\xi = \dot{\chi}_{\xi'}$. $\blacksquare$

Lemma 2.15 (Product Lemma). *Let M be a transitive model of ZFC, $\mathbb{P}, \mathbb{Q}$ be forcing posets in M .*

(1) If $G \subseteq \mathbb{P}$ is $\mathbb{P}$ -generic over M , $H \subseteq \mathbb{Q}$ is $\mathbb{Q}$ -generic over $M[G]$ then $G \times H$ is a $\mathbb{P} \times \mathbb{Q}$ -generic filter over M .

(2) If $K \subseteq \mathbb{P} \times \mathbb{Q}$ is $\mathbb{P} \times \mathbb{Q}$ -generic over M , set $G := \{p \in \mathbb{P} : \exists q \in \mathbb{Q}, (p, q) \in K\}$ and $H := \{q \in \mathbb{Q} : \exists p \in \mathbb{P}, (p, q) \in K\}$ then G is $\mathbb{P}$ -generic over M , H is $\mathbb{Q}$ -generic over $M[G]$.

Proof. (1) Let $D \subset \mathbb{P} \times \mathbb{Q}, D \in M$ be dense. Define $D' = \{q \in \mathbb{Q} : \exists p \in G, (p, q) \in D\}$, clearly $D' \in M[G]$. We claim that D' is dense in $\mathbb{Q}$: Fix $r \in \mathbb{Q}$, we want to find $q \in D', q \leq r$. Define $E_r = \{p \in \mathbb{P} : \exists q \in \mathbb{Q}, q \leq r, (p, q) \in D\}$. For any $s \in \mathbb{P}, (s, r) \in \mathbb{P} \times \mathbb{Q}$ thus $\exists (p, q) \in D, (p, q) \leq (s, r)$ thus $p \leq s$, so E_r is dense in $\mathbb{P}$. Say $p \in G \cap E_r$ then there exists $q \in \mathbb{Q}, q \leq r, (p, q) \in D$. Hence D' is dense in $\mathbb{Q}$. And then $H \cap D' \neq \emptyset$, gives $G \times H \cap D \neq \emptyset$.

(2) Let $D \subset \mathbb{P}, D \in M$ be dense. Take $D \times \mathbb{Q}$, this is clearly dense in $\mathbb{P} \times \mathbb{Q}$ thus $G \cap D \neq \emptyset$.

Now let $D \subset \mathbb{Q}, D \in M[G]$ be dense. Let $\tau^G = D$ and there is $p^* \in G, p^* \Vdash \tau$ is dense in $\check{\mathbb{Q}}$. Define $D^* = \{(p, q) \in \mathbb{P} \times \mathbb{Q} : p \leq p^* \wedge p \Vdash \check{q} \in \tau\}$. We claim that D^* is dense below $(p^*, 1_{\mathbb{Q}})$.

Take any $(p, q) \leq (p^*, 1_{\mathbb{Q}})$ since $p \Vdash \tau$ is dense in $\check{\mathbb{Q}}$ thus $p \Vdash \exists \sigma \in \check{\mathbb{Q}} (\sigma \in \tau) \wedge (\sigma \leq \check{q})$. Then by forcing theorem the set $\{s \leq p : s \Vdash \exists t \in \mathbb{Q} (\check{t} \in \tau) \wedge (\check{t} \leq \check{q})\}$ is dense below p . Thus there exists $s \leq p, t \in \mathbb{Q}$ such that $s \Vdash \check{t} \in \check{\mathbb{Q}} \wedge \check{t} \leq \check{q}$ and thus $(s, t) \in D^*$. ■

Warning: If g is $\mathbb{P}$ -generic over M and h is $\mathbb{Q}$ -generic over M then $g \times h$ is not necessarily a $\mathbb{P} \times \mathbb{Q}$ -generic filter over M . See Sheet 6.

By using the product lemma, in fact the following is true:

$$\mathbb{C}(\alpha) \cong \begin{cases} \mathbb{C}(\alpha) \times \mathbb{C} & \text{if } \alpha \geq \omega \\ \mathbb{C}(\alpha - 1) \times \mathbb{C} & \text{if } \alpha < \omega \end{cases}$$

and then g_{ξ} is $\mathbb{C}$ -generic over $M[(g_{\gamma} : \gamma < \alpha, \gamma \neq \xi)], M[g] = M[(g_{\gamma} : \gamma < \alpha, \gamma \neq \xi)][g_{\xi}]$, then χ_{ξ} is not definable over $M[g]$ from parameters in $M[(g_{\gamma} : \gamma < \alpha, \gamma \neq \xi)]$ by the previous lemma.

Theorem 2.16. Let M be a transitive model of ZFC, let $\alpha \in M, g$ be a $\mathbb{C}(\alpha)$ -generic filter over M and $x \in {}^{\omega}\omega \cap M[g]$. There is then some $h \in M[g]$, here h is $\mathbb{C}$ -generic over M with $x \in M[h]$.

Proof. Let $x = \tau^g$ for every pair $(n, m) \in \omega \times \omega$, let $A_{n,m}$ be an maximal antichain of $p \in \mathbb{C}(\alpha)$ with $p \Vdash \tau(\check{n}) = \check{m}$, since $\mathbb{C}(\alpha)$ has the c.c.c. $A_{n,m}$ is countable. Let $\bar{\tau} = \{((n, \check{m}), p) : p \in A_{n,m} ; n, m < \omega\}$ then $\bar{\tau}^g = \tau^g$. Let $X := \bigcup \{\text{supp}(p) : \exists (n, m) \in \omega \times \omega \text{ with } p \in A_{n,m}\}, \bar{\bar{X}} \leq \omega$. Since $\mathbb{C}(\alpha) \cong \mathbb{C}(X) \times \mathbb{C}(\alpha \setminus X)$, but $\mathbb{C}(X) \cong \mathbb{C}(\omega) \cong \mathbb{C}$. Here $\mathbb{C}(X) = \{p \in {}^X\mathbb{C} : \overline{\text{supp}(p)} < \aleph_0\}$ By the choice of $X, x = \bar{\tau}^g \in M[g \restriction_X]$ therefore $x \in M[h]$ where $h \in M[g]$ is $\mathbb{C}$ -generic, i.e., we rewrite $g \restriction_X$ as h via an isomorphism $\pi : \mathbb{C}(X) \rightarrow \mathbb{C}$. ■

Theorem 2.17. Let M be a transitive model of ZFC and $\alpha \in M, \alpha \geq \omega_1^M$ and let g be $\mathbb{C}(\alpha)$ -generic over M . Then in $M[g]$ there is no well-order of the reals which are definable from ordinals and reals, i.e., there is no formula φ and no ordinals $\alpha_0, \dots, \alpha_{n-1} \in M[g]$ and reals $x_0, \dots, x_{m-1} \in M[g]$ such that $\{(x, y) \in ({}^{\omega}\omega \times {}^{\omega}\omega) \cap M[g] : M[g] \models \varphi(x, y, \alpha_0, \dots, \alpha_{n-1}, x_0, \dots, x_{m-1})\}$ is a well-order of ${}^{\omega}\omega \cap M[g]$.

Proof. Suppose there exists a formula φ and ordinals $\alpha_0, \dots, \alpha_{n-1} \in M[g]$ and reals $x_0, \dots, x_{m-1} \in M[g]$ such that $\{(x, y) \in (\omega^\omega \times \omega^\omega) \cap M[g] : M[g] \models \varphi(x, y, \alpha_0, \dots, \alpha_{n-1}, x_0, \dots, x_{m-1})\}$ is a well-order of $\omega^\omega \cap M[g]$. By what we proved earlier, we may pick a (large enough) set $X \subseteq \alpha, X \in M$ which is countable in M such that $x_0, \dots, x_{m-1} \in M[g \upharpoonright_X] \subseteq M[g] = M[g \upharpoonright_X][g \upharpoonright_{(\alpha \setminus X)}], \mathbb{C}(X) \times \mathbb{C}(\alpha \setminus X) \cong \mathbb{C}(\alpha)$. Let $\beta \in \alpha \setminus X$ and look at $g \upharpoonright_{\{\beta\}}$ which may be constructed as a Cohen real. Cohen real over $M[g \upharpoonright_X]$ in particular $\bigcup g \upharpoonright_{\{\beta\}}$ is not definable in $M[g]$ from parameters in $M[g \upharpoonright_X]$. By the hypothesis there is some ordinal $\gamma \in M$ such that from the point of view of $M[g], \bigcup g \upharpoonright_{\{\beta\}}$ is the γ^{th} -real in the well-order defined by $\varphi(-, -, \alpha_0, \dots, \alpha_{n-1}, x_0, \dots, x_{m-1})$. In particular $\bigcup g \upharpoonright_{\{\beta\}}$ would be definable in $M[g]$ from parameters in $M[g \upharpoonright_X]$, contradiction! ■

Corollary 2.18. *Let M be a transitive model of ZFC and $\alpha \geq \omega_1^M, g$ be a $\mathbb{C}(\alpha)$ -generic filter over M . Let $\mathbb{R}^* = \omega^\omega \cap M[g]$ and $N = (\text{HOD}_{\mathbb{R}^*})^{M[g]}$. Then $N \models \text{"ZF and there is no well-order of the reals"}$.*

Proof. If N had a well-order, say $\leq$, then $\leq$ would be definable inside $M[g]$ using finitely many ordinals and reals from $\mathbb{R}^*$, contradiction! ■

2.1.2 A model of ZF in which there is a countable set of countable sets of reals whose union is uncountable

Definition 2.19. *Let X be a set and let α be an ordinal. Define $\text{Col}(\alpha, X) := \{p : \exists \beta < \alpha, p \in {}^\beta X\}$, ordered by $p \leq q$ if and only if $p \supseteq q$. To force CH, use $\text{Col}(\omega_1, 2^{\aleph_0})$.*

From now, we are going to make use only of $\text{Col}(\omega, \gamma)$ for some ordinals γ .

Definition 2.20. *Let β be an ordinal, then*

$$\text{Col}(\omega, < \beta) := \{p : (p \text{ is a function with } \text{dom}(p) = \beta) \wedge (\forall \gamma < \beta, p(\gamma) \in \text{Col}(\omega, \gamma)) \wedge \overline{\overline{\text{supp}(p)}} < \aleph_0\}$$

ordered by $p \leq q \iff p(\gamma) \leq_{\text{Col}(\omega, \gamma)} q(\gamma)$ for all γ .

Lemma 2.21. *Let M be a transitive model of ZFC and let $X \in M$, if g is $\text{Col}(\omega, X)$ -generic over M then X is at most countable in $M[g]$.*

Proof. For any $x \in X$, define $D_x := \{p \in \text{Col}(\omega, X) : \exists n < \omega, p(n) = x\}$, $D_x \in M$ and D_x is dense, hence $D_x \cap g \neq \emptyset$. Let $f = \bigcup g \in M[g]$, f is a function from ω to X and by the proceeding paragraph f is actually onto. ■

Definition 2.22. *Let κ be a cardinal, then*

$$\text{Col}(\omega, < \kappa) := \{p : (p \text{ is a function with } \text{dom}(p) = \kappa) \wedge (\forall \xi < \kappa, p(\xi) \in \text{Col}(\omega, \xi)) \wedge \overline{\overline{\text{supp}(p)}} < \aleph_0\}$$

ordered by $p \leq q \iff p(\xi) \leq_{\text{Col}(\omega, \xi)} q(\xi)$ for all $\xi < \kappa$.

Notice that $\overline{\overline{\text{Col}(\omega, < \kappa)}} = \kappa$.

Lemma 2.23. *Let M be a transitive model of ZFC and let $\kappa \in M$, if g is $\text{Col}(\omega, < \kappa)$ -generic over M then $M[g] \models \overline{\overline{\xi}} \leq \aleph_0$ for all $\xi < \kappa$.*

Proof. Directly follows from the previous lemma and the definition. ■

Definition 2.24. $\text{Col}^*(\omega, Y) = \{p: (p \text{ is a function with } \text{dom}(p) = Y) \wedge (\forall X \in Y, p(X) \in \text{Col}(\omega, X)) \wedge (\overline{\text{supp}(p)} < \aleph_0)\}$, ordered as before.

E.g. for all $Y \subseteq \kappa, Y \neq \emptyset, \text{Col}(\omega, < \kappa) \cong \text{Col}^*(\omega, Y) \times \text{Col}^*(\omega, \kappa \setminus Y)$. In particular, this gives a proof of the last lemma immediately: $\text{Col}(\omega, < \kappa) \cong \text{Col}^*(\omega, \{\xi\}) \times \text{Col}^*(\omega, \kappa \setminus \{\xi\}), \text{Col}(\omega, \xi) = \text{Col}^*(\omega, \{\xi\})$.

Note that if g is a $\text{Col}(\omega, < \kappa)$ -generic filter over M then $M[g] \models \bar{\kappa} \leq \aleph_1$, additionally if κ is singular then $M[g] \models \bar{\kappa} = \aleph_0$ as $\text{cf}(\kappa) = \omega$ in $M[g]$, follows from 2.23.

Later we want to study $\text{Col}(\omega, \kappa)$ for regular limit uncountable cardinal (they are called weakly inaccessible cardinal), for now we want to study what happens if we force with $\text{Col}(\omega, < \aleph_\omega)$.

Lemma 2.25. *Let M be a transitive model of ZFC and let $\mathbb{P} \in M$ be a poset such that $M \Vdash \text{"}\mathbb{P} \text{ has the } \kappa\text{-c.c."}$ for some $\kappa \in M$ which is an uncountable cardinal of M . If g is $\mathbb{P}$ -generic over M then $M[g]$ and M has the same cardinal $\geq \kappa$.*

Proof. Suppose not then let $\lambda \geq \kappa$ be the least such that $M \models \text{"}\lambda \text{ is a cardinal"}$ and $M[g] \models \text{"}\lambda \text{ is not a cardinal"}$. Then $\lambda = (\eta^+)^M$ for some $\eta < \lambda, M[g] \models \exists f (f: \eta \rightarrow \lambda \text{ is onto})$. Say $f = \tau^g$ where $\tau \in M^{\mathbb{P}}$.

Let $p \in g, p \Vdash \text{"}\tau: \check{\eta} \rightarrow \check{\lambda} \text{ is onto."}$ For every $\xi < \eta$, let $X_\xi = \{\nu < \lambda: \exists q \leq p (q \Vdash \tau(\check{\xi}) = \check{\nu})\}$. For each $\nu \in X_\xi$, pick $q_\nu \leq p$ such that $q_\nu \Vdash \tau(\check{\xi}) = \check{\nu}$. Then $\{q_\nu: \nu \in X_\xi\}$ is an antichain as $\nu \neq \nu', q_\nu \perp q_{\nu'}$. Hence $\overline{X_\xi} = \overline{\{q_\nu: \nu \in X_\xi\}} < \kappa$, i.e., $M \models \overline{X_\xi} < \kappa$.

Case 1: $\kappa \leq \eta$

We define $F: \eta \times \kappa \rightarrow \lambda$ in M as follows: $\forall \xi < \eta$, let $f_\xi: \kappa \rightarrow X_\xi$ be onto, for $(\xi, i) \in \eta \times \kappa$, let $F((\xi, i)) = f_\xi(i)$. We claim F is onto: Let $\nu < \lambda$, there is some $\xi < \eta$ such that $f(\xi) = \nu$ since f is onto, then $\nu \in X_\xi$ as there is some $q \leq p, q \in g$ with $q \Vdash \tau(\check{\xi}) = \check{\nu}$. Hence $\nu = f_\xi(i)$ for some $i < \kappa$. Therefore $\nu \in \text{ran}(F)$, but then $F \in M, \lambda \leq \overline{\eta \times \kappa}$ and contradict to the fact $\kappa \leq \eta < \lambda$.

Case 2: $\kappa = \lambda = \eta^+$

In $M, \kappa = \lambda = \eta^+$, so κ is regular. Since $f: \eta \rightarrow \lambda = \kappa$ is onto so it is also cofinal function.

Define in $M, h: \eta \rightarrow \kappa$ by $h(\xi) = \sup X_\xi$. By the regularity of κ , $h(\xi) < \kappa$ for all $\xi < \eta$, so h is well-defined. Now we want to show h is a cofinal function in κ . Take $\beta < \kappa$, since $M[g] \models \text{ran}(f)$ is cofinal in κ , there exists $\xi < \eta$ such that $f(\xi) > \beta$. Let $q \in g, q \leq p$ be such that $q \Vdash \tau(\check{\xi}) > \check{\beta}$, then $f(\xi) \in X_\xi$. It looks a bit weird because it is an abuse of notation. Thus $f(\xi) < \sup X_\xi = h(\xi)$, hence $h(\xi) > \beta$. So $h \in M$ is a cofinal function but $\eta < \text{cf}(\kappa) = \kappa$ and then leads to a contradiction. ■

Lemma 2.26. *Let M be a transitive model of ZFC and let g be $\text{Col}(\omega, < \aleph_\omega^M)$ -generic over M . Then $\aleph_{\omega+\alpha}^M = \aleph_\alpha^{M[g]}$ for all $\alpha \geq 1$. More precisely, $\overline{\aleph_\omega^M} = \aleph_0^{M[g]}$ and λ is a cardinal in $M[g]$ for all M -cardinals $\lambda \geq \aleph_{\omega+1}^M$.*

Now let us fix M to be a transitive model of ZFC and $g \subseteq \text{Col}(\omega, < \aleph_\omega^M)$ a M -generic filter.

Definition 2.27. *For a cardinal $\kappa, H_\kappa := \{x: \overline{\text{TC}(\{x\})} < \kappa\}$, hereditarily finite $\text{HF} = H_{\aleph_0}$, hereditarily countable $\text{HC} = H_{\aleph_1}$.*

Let $\mathbb{R}^* = \text{HC}^* = \bigcup \{ \text{HC}^{M[g \upharpoonright \alpha]} : \alpha < \aleph_\omega^M \}; \forall \alpha < \aleph_\omega^M, \text{Col}(\omega, < \aleph_\omega^M) \cong \text{Col}^*(\omega, \alpha) \times \text{Col}^*(\omega, \aleph_\omega^M \setminus \alpha)$. By way of notation, $g \upharpoonright_\alpha = \{p \upharpoonright_\alpha : p \in g\}$, $g \upharpoonright_{(\aleph_\omega^M \setminus \alpha)} = \{p \upharpoonright_{(\aleph_\omega^M \setminus \alpha)} : p \in g\}$, then $g \upharpoonright_\alpha$ is $\text{Col}(\omega, \alpha)$ -generic over M and $g \upharpoonright_{(\aleph_\omega^M \setminus \alpha)}$ is $\text{Col}^*(\omega, \aleph_\omega^M \setminus \alpha)$ -generic over $M[g \upharpoonright_\alpha]$

Proposition 2.28 (Key Claim). *Let $N = \text{HOD}_{\text{HC}^* \cup \{\text{HC}^*, (\aleph_n^M : n < \omega)\}}^{M[g]}$, then $N \models \text{ZF} \wedge \omega_1 = \aleph_\omega^M$.*

In particular, ω_1 has countable cofinality in N as $(\aleph_n^M : n < \omega) \in N$.

We will be there if we once proved: if $x \in \text{HC}^N$ then $x \in \text{HC}^*$, i.e., $x \in \text{HC}^{M[g \upharpoonright \alpha]}$ for some $\alpha < \aleph_\omega^M$. Now let us show why this statement implies $\omega_1^N = \aleph_\omega^M$.

Lemma 2.29. *Suppose $\forall x \in \text{HC}^N, x \in \text{HC}^*$. Then $\omega_1^N = \aleph_\omega^M$.*

Proof. $\omega_1^N \geq \aleph_\omega^M$: If $\alpha < \aleph_\omega^M$, let $\xi = \alpha + 1$ and consider $\text{HC}^{M[g \upharpoonright \xi]} \subset \text{HC}^* \subset N$. Since $\text{HC}^{M[g \upharpoonright \xi]} \models "$ α is countable", thus $\aleph_\omega^M \leq \omega_1^N$.

$\omega_1^N \leq \aleph_\omega^M$: Suppose in N there is some $f : \omega \rightarrow \aleph_\omega^M$ onto. Then $f \in \text{HC}^N$ therefore by assumption $f \in \text{HC}^*$. Hence there is some $\xi < \aleph_\omega^M$ with $f \in M[g \upharpoonright \xi]$. But since $g \upharpoonright_\xi$ is $\text{Col}(\omega, < \xi)$ -generic over M and $M \models \overline{\text{Col}(\omega, < \xi)} = \bar{\xi}$ so $\text{Col}(\omega, < \xi)$ has the $\bar{\xi}^+$ -c.c. in M and all M -cardinals $\geq \bar{\xi}^{+M}$ are preserved in $M[g \upharpoonright \xi]$. But $\xi < \aleph_\omega^M$ so $\aleph_\omega^M$ is still a cardinal in $M[g \upharpoonright \xi]$ which leads to a contradiction. ■

Since $N = (\text{HOD}_{\text{HC}^* \cup \{\text{HC}^*, (\aleph_n^M : n < \omega)\}})^{M[g]}$ so $\text{HC}^* \subset \text{HC}^N$ holds trivially.

Now let us prove $\forall x \in \text{HC}^N \rightarrow x \in \text{HC}^*$.

Lemma 2.30. *If $x \in \text{HC}^N$ then $x \in \text{HC}^*$.*

Proof. Let us fix $x \in \text{HC}^N$. Inside N , let $f : \omega \rightarrow \text{TC}(\{x\})$ be bijective. Let $E \subseteq \omega \times \omega$ be such that $(n, m) \in E$ if and only if $f(n) \in f(m)$, i.e., $(\omega; E) \stackrel{f}{\cong} (\text{TC}(\{x\}); \in)$. It suffices to prove $E \in \text{HC}^*$ because then if $E \in \text{HC}^{M[g \upharpoonright \xi]}, \xi < \aleph_\omega^M$ and then $f, \text{TC}(\{x\}) \in M[g \upharpoonright \xi]$ (by Axiom Schema of Replacement and Separation), hence $x \in \text{HC}^{M[g \upharpoonright \xi]} \subset \text{HC}^*$.

Let us prove $E \in \text{HC}^*$. Since $E \in N$ so $(n, m) \in E$ if and only if $M[g] \models \varphi(n, m, x_1, \dots, x_k, \text{HC}^*, (\aleph_n^M : n < \omega), \alpha_1, \dots, \alpha_l)$ for some formula φ , some elements $x_1, \dots, x_k \in \text{HC}^*$ and ordinals $\alpha_1, \dots, \alpha_l$. Let us pick $\xi < \aleph_\omega^M$ such that $x_1, \dots, x_k \in \text{HC}^{M[g \upharpoonright \xi]}$. Then

$$(n, m) \in E \iff M[g \upharpoonright \xi][g \upharpoonright_{[\xi, \aleph_\omega^M]}] \models \varphi(n, m, x_1, \dots, x_k, \text{HC}^*, (\aleph_n^M : n < \omega), \alpha_1, \dots, \alpha_l)$$

Notice that HC^* is the only parameter which does not lie in $M[g \upharpoonright \xi]$.

In fact, $\text{Col}^*(\omega, X)$ is homogeneous for every X . Therefore if in the above definition of E the parameter HC^* does not show up then we could argue as follows: $E \subseteq \omega \times \omega$ is definable in $M[g] = M[g \upharpoonright \xi][g \upharpoonright_{[\xi, \aleph_\omega^M]}]$ from parameters in $M[g \upharpoonright \xi]$, but $M[g]$ is a $\text{Col}^*(\omega, [\xi, \aleph_\omega^M])$ -generic extension of $M[g \upharpoonright \xi]$, hence by homogeneity, $E \in M[g \upharpoonright \xi]$, i.e., $E \in \text{HC}^{M[g \upharpoonright \xi]} \subset \text{HC}^*$.

From now on, let us assume $\xi = 0$ for convenience.

Claim: There is some $\tau \in M^{\text{Col}(\omega, < \aleph_\omega^M)}$ such that $\tau^g = \text{HC}^*$ and for all $p, q \in \text{Col}(\omega, < \aleph_\omega^M)$ there is an isomorphism π such that $\pi(p) \parallel q$ and $\tilde{\pi}(\tau) = \{(\tilde{\pi}(\sigma), \pi(r)) : (\sigma, r) \in \tau\} = \tau$.

Proof of the claim. Let us define a name for $\text{HC}^* \cap \mathcal{P}(\omega)$ with the above properties, i.e.,

$$\tau = \{(\sigma, p) : \exists \xi < \aleph_\omega^M (p \in \text{Col}(\omega, < \xi)) \wedge (\text{the elements of } \sigma \text{ are all pairs } (\check{n}, q), \text{ where } n < \omega, \text{supp}(q) \subset \xi)\}.$$

In order to define τ , let us introduce the following notation: Let $\xi < \aleph_\omega^M, p \in \text{Col}(\omega, < \xi)$, we write p^* for the element in $\text{Col}(\omega, < \aleph_\omega^M)$ such that $p^* \restriction_\xi = p$ and $p^*(i) = \mathbb{1}_{\text{Col}(\omega, i)}$ for $\xi \leq i < \aleph_\omega^M$, we may also write $\tau^* = \{(\sigma^*, p^*) : (\sigma, p) \in \tau\}$ for any $\text{Col}(\omega, < \xi)$ -name τ then $\tau^{*g} = \tau^{g \restriction_\xi}$.

Every $x \subseteq \omega$ in $M[g \restriction_\xi]$ has a name of the form $\{(\check{n}, p) : n < \omega, p \in A_n\}$ where $A_n \subseteq \text{Col}(\omega, < \xi)$.

Take $x = \tau^{g \restriction_\xi}$ and let $\sigma = \{(\check{n}, p) : n < \omega \wedge p \in \text{Col}(\omega, < \xi) \wedge p \Vdash \check{n} \in \tau\}$.

Let $\tau = \bigcup_{\xi < \aleph_\omega^M} \tau_\xi = \bigcup_{\xi < \aleph_\omega^M} \{(\sigma, p^*) : \sigma \in M^{\text{Col}(\omega, < \xi)} \text{ is a name for a subset of } \omega \wedge p \in \text{Col}(\omega, < \xi)\}$. For a fixed ξ , τ_ξ is a name for $\mathcal{P}(\omega) \cap \text{HC}^{M[g \restriction_\xi]}$ then $(\tau_\xi)^*$ is a name for $\mathcal{P}(\omega) \cap \text{HC}^{M[g \restriction_\xi]}$ in $M^{\text{Col}(\omega, < \aleph_\omega^M)}$. Hence $\tau^* = \bigcup_{\xi < \aleph_\omega^M} (\tau_\xi)^*$ is a name for $\bigcup_{\xi < \aleph_\omega^M} \mathcal{P}(\omega) \cap M[g \restriction_\xi] = \mathcal{P}(\omega) \cap \text{HC}^*$. ■

Finally, we are left with having to prove: if $p, q \in \text{Col}(\omega, < \aleph_\omega^M)$ there is then an isomorphism π with $\pi(p) \parallel q$ and $\tilde{\pi}(\tau) = \tau$. Let p, q be given, extend p, q in a way to p', q' such that $\text{supp}(p') = \text{supp}(q')$ and $\text{dom}(p'(i)) = \text{dom}(q'(i))$ for all $i < \aleph_\omega^M$. As for Cohen forcing, for each i there is an isomorphism $\pi_i : \text{Col}(\omega, i) \rightarrow \text{Col}(\omega, i)$ with $\pi_i(p'(i)) \parallel q'(i)$. If $i \notin \text{supp}(p')$ then put $\pi_i = \text{id}$, also if $i \in \text{supp}(p'), n \geq \text{dom}(p'(i))$ then put $(\pi_i(r))(n) = r(n)$ for all $r \in \text{Col}(\omega, i)$ with $n \in \text{dom}(r)$. For $r \in \text{Col}(\omega, < \aleph_\omega^M)$, $(\pi(r))(i) = \pi_i(r(i))$ and it is straightforward to verify $\tilde{\pi}(\tau) = \tau$ for all $\tau \in M^{\text{Col}(\omega, < \aleph_\omega^M)}$. ■

In the model N which we constructed, $\mathbb{R}$ is the countable union of countable sets of reals as $\mathbb{R}^N = \bigcup \{\mathbb{R}^{M[g \restriction_{\aleph_n^M}]} : n < \omega\}$ and $N \models \mathbb{R}$ is uncountable!

2.2 Solovay's model: A model of ZF where every set of real is Lebesgue measurable and has the Baire property and perfect set property

Lemma 2.31. *Let $\kappa \geq \aleph_1$ be a regular cardinal. Then $\text{Col}(\omega, < \kappa)$ has the κ -c.c.*

Remark: If κ is singular or in particular $\aleph_\omega$ then $\text{Col}(\omega, < \kappa)$ does not have the κ -c.c. as forcing with it collapse κ .

Proof. Let $\{p_i : i < \kappa\} \subset \text{Col}(\omega, < \kappa)$. We may assume that there is $n_0 < \omega$ such that $\overline{\overline{\text{supp}(p_i)}} = n_0$ for all $i < \kappa$, write $\text{supp}(p_i) = \{\xi_1^i, \dots, \xi_{n_0}^i\}$ such that $\xi_1^i < \xi_2^i < \dots < \xi_{n_0}^i$.

Case 1: If there exists $\xi < \kappa$ with $\{i < \kappa : \xi_{n_0}^i \geq \xi\}$ has size $< \kappa$. Then we may assume $\text{supp}(p_i) \subset \xi$ for all $i < \kappa$, then $p_i = (p_i \restriction_\xi)^*$ (same as how we defined in the proof of 2.30). Since $\text{Col}(\omega, < \xi)$ has cardinality $\bar{\xi}$, thus $\{p_i : i < \kappa\}$ can not be an antichain in κ .

Case 2: If the set $\{\xi_{n_0}^i : i < \kappa\}$ is unbounded in κ . Let $m_0 \leq n_0$ be the least n such that $\{\xi_{m_0}^i : i < \kappa\}$ is unbounded in κ . Note that if $m_0 = 1$ then $\{p_i : i < \kappa\}$ can not be an antichain: let p_1 be any condition in the sequence. Since for all $n, 1 \leq n \leq n_0$ the set $\{\xi_n^i : i < \kappa\}$ is unbounded in κ so we can find p_2 with $\xi_1^2 > \xi_{n_0}^1$ then $\text{supp}(p) \cap \text{supp}(q) = \emptyset$ thus $p \parallel q$.

Now we may assume $m_0 > 1$, then there is some $\xi < \kappa$ with $\xi_{m_0-1}^i < \xi$ for all $i < \kappa$ as $\{\xi_{m_0-1}^i : i < \kappa\}$ is bounded in κ . Since $\text{Col}(\omega, < \xi)$ has cardinality $\bar{\xi}$, then $\overline{\overline{\{p_i \restriction_\xi : i < \kappa\}}} = \bar{\xi}$. So κ many of them are pairwise equal, by regularity, $p_i \restriction_\xi = p_j \restriction_\xi$ for all $i, j < \kappa$. Then by using the same argument in the case $m_0 = 1$, we can show that $\{p_i : i < \kappa\}$ can not be an antichain. ■

Remark: $\text{Col}(\omega, < \aleph_1) \cong \text{add } \aleph_1 \text{ many Cohen reals} = \mathbb{C}(\omega_1)$.

Question(by Hausdorff): Is there an uncountable regular limit cardinal?

Definition 2.32. A cardinal is called **weakly inaccessible** if it is an uncountable regular limit cardinal.

An uncountable regular cardinal κ is called **(strongly) inaccessible** if $\forall \mu < \kappa, 2^\mu < \kappa$.

Since in ZFC we have $\mu^+ \leq 2^\mu$ for all infinite cardinal μ , so if $\forall \mu < \kappa, 2^\mu < \kappa$ then $\mu^+ < \kappa$ and thus κ is a limit cardinal. Hence all (strongly) inaccessible cardinals are also weakly inaccessible.

If GCH holds true, i.e., $\aleph_{i+1} = 2^{\aleph_i}$ for all $i < \omega$ then every weakly inaccessible cardinal is also (strongly) inaccessible.

Theorem 2.33. Let κ be strongly inaccessible, then $(V_\kappa, \epsilon) \models \text{ZFC}$. In particular, ZFC can not prove the existence of a strongly inaccessible cardinal.

Proof. V_κ is transitive, so both (Ext) and (Fund) hold. As $\kappa > \omega$, (Inf) holds in V_κ . Since κ is a limit cardinal so V_κ is closed under $x, y \mapsto \{x, y\}$ as well as $x \mapsto \bigcup x$, hence $V_\kappa \vdash (\text{Pair}), (\text{Union})$.

Let $x \in V_\kappa$, since κ is limit, so there exists $\alpha < \kappa$ such that $x \in V_\alpha$ then $\mathcal{P}(x) \in V_{\alpha+1} \subset V_\kappa$. Hence (Pow) holds true. If $x \in V_\kappa$ then any subset of x also lie in V_κ hence $V_\kappa \models (\text{AC})$ and (Aus).

As for (Rep), we first prove $H_\kappa = V_\kappa$. For $V_\kappa \subseteq H_\kappa$, it suffices to show that $V_\alpha \subseteq H_\kappa$ for all $\alpha < \kappa$. By induction, let $\alpha < \kappa$ be such that $V_\beta \in H_\kappa$ for all $\beta < \alpha$. If $\alpha = \beta + 1$, say $\overline{\overline{V_\beta}} = \mu < \kappa$ then $\overline{\overline{V_{\beta+1}}} = 2^{\overline{\overline{V_\beta}}} = 2^\mu < \kappa$ thus $V_\alpha \in H_\kappa$. If α is a limit, then $V_\alpha = \bigcup_{\beta < \alpha} V_\beta$ also has size $< \kappa$ as $\overline{\overline{V_\beta}} < \kappa, \overline{\overline{\alpha}} < \kappa$ and $\text{cf}(\kappa) = \kappa$. For $H_\kappa \subseteq V_\kappa$, by ϵ -induction, let $x \in H_\kappa$ and assume inductively that $x \subset V_\kappa$. For any $y \in x$, let α_y be the least ordinal such that $y \in V_{\alpha_y}$. Since $\{\alpha_y : y \in x\}$ is bounded in κ as $\text{TC}(\{x\})$ has cardinality $< \kappa$, let $\gamma < \kappa$ be such that $\alpha_y < \gamma$ for all $y \in x$. Then $x \in V_{\gamma+1} \subset V_\kappa$ so $x \in V_\kappa$.

Now, let $F: V_\kappa \rightarrow V_\kappa$ (it does not even necessarily to be definable over V_κ) be a function and $x \in V_\kappa$. We want to show $F''x \in V_\kappa$. Since $F''x$ has cardinality $< \kappa$ and $\overline{\overline{\text{TC}(\{F''a\})}} < \kappa$ for all $a \in x$. Therefore $\text{TC}(\{F''x\}) = \{F''x\} \cup \bigcup_{a \in x} \text{TC}(\{F''a\})$ and this has size $< \kappa$ by regularity of κ . And this gives (Rep) in V_κ . ■

Remark: The same argument shows that $V_\omega = H_\omega = \text{HF} \models \text{ZFC} - (\text{Inf})$.

We now force over a transitive model M of $\text{ZFC} + "$ κ is an inaccessible cardinal" with the forcing $\text{Col}(\omega, < \kappa)$. Let g be a $\text{Col}(\omega, < \kappa)$ -generic filter over M , then $\omega_1^{M[g]} = \kappa^M$. Let us fix this situation for now.

Lemma 2.34. Let $x \in {}^\omega \omega \cap M[g]$, there is then some $\alpha < \kappa$ such that $x \in M[g \restriction_\alpha]$ and κ is inaccessible in $M[g \restriction_\alpha]$.

Proof. Instead of proving $x \in {}^\omega \omega$, we prove it for $x \in \mathcal{P}(\omega)$. As before, there is some $\tau \in M^{\text{Col}(\omega, < \kappa)}$ such that $\tau = \{(\tilde{n}, p) : n < \omega \wedge p \in A_n \wedge p \Vdash \tilde{n} \in \sigma\}$ where each $A_n \subseteq \text{Col}(\omega, < \kappa)$ is an maximal antichain of $q \in \text{Col}(\omega, < \kappa)$ deciding $\tilde{n} \in \tau$.

To be more precise, let $x = \sigma^g$, for each $n < \omega$, pick A_n , an maximal antichain in $\{q : q \Vdash \tilde{n} \in \sigma \vee q \Vdash \neg(\tilde{n} \in \sigma)\}$, i.e., the set of forcing condition who decides $\tilde{n} \in \sigma$ and let $\tau = \{(\tilde{n}, p) : n < \omega \wedge p \in A_n \wedge p \Vdash$

$\check{n} \in \sigma\}$, then $\tau^g = \sigma^g$. For $\sigma^g \subseteq \tau^g$: if $n \in x$ then there exists some $p \in g$ such that $p \Vdash \check{n} \in \sigma$, since $g \cap A_n \neq \emptyset$, say $\bar{p} \in g \cap A_n$, $p, \bar{p}$ both lie in g so they are compatible, thus $\bar{p} \Vdash \check{n} \in \sigma$ then $(\check{n}, \bar{p}) \in \tau$ and $n \in \tau^g$. For $\tau^g \subseteq \sigma^g$: It is clear by checking the definition.

Since $\text{supp}(p)$ is a finite subset of κ for each $p \in \text{Col}(\omega, < \kappa)$ and $\overline{\bigcup_{n < \omega} A_n} < \kappa$, so there exists some $\alpha < \kappa$ such that $\text{supp}(p) \subset \alpha$ for all $p \in \bigcup_{n < \omega} A_n$.

Let $\bar{\tau} = \{(\check{n}, p \restriction_\alpha) : n < \omega \wedge p \in A_n \wedge p \Vdash \check{n} \in \sigma\}$ then $(\bar{\tau})^{g \restriction_\alpha} = \tau^g$. Hence $x \in M[g \restriction_\alpha]$, as $M \models \overline{\text{Col}(\omega, < \alpha)} = \bar{\alpha} < \kappa$ so $\text{Col}(\omega, < \alpha)$ has the $\bar{\alpha}^+$ -c.c. then M and $M[g]$ have the same cardinal $\geq \bar{\alpha}^+$. It left to verify $M[g \restriction_\alpha] \models \kappa$ is inaccessible, we need to show $M[g \restriction_\alpha] \models \forall \mu < \kappa, 2^\mu < \kappa$. Now fix $\mu < \kappa$, if $x \in \mathcal{P}(\mu) \cap M[g \restriction_\alpha]$ then $x = \tau^{g \restriction_\alpha}$ where $\tau \in M, \tau = \{(\check{\xi}, p) : \xi < \mu, p \in A_\xi, p \Vdash \check{\xi} \in \sigma\}$ where $\overline{A_\xi} \leq \bar{\alpha}$ for each ξ . Inside M there are only $(2^\alpha)^\mu$ many such names τ and hence in $M[g \restriction_\alpha]$ there are only $< \kappa$ many subsets of μ , this shows $M[g \restriction_\alpha] \models \forall \mu < \kappa, 2^\mu < \kappa$. ■

The very same argument shows that if $f \in {}^\omega \text{Ord} \cap M[g]$ then there is some $\xi < \kappa$ with $f \in M[g \restriction_\xi]$.

Now let $N = \text{HOD}_{M \cup ({}^\omega \text{Ord} \cap M[g])}^{M[g]}$, i.e., $x \in N$ if and only if every element of $\text{TC}(\{x\})$ is definable inside $M[g]$ with parameters in $M \cup ({}^\omega \text{Ord} \cap M[g])$, it is clear that $N \models \text{ZF}$.

Recall (Notation): We construct $\mathbb{R} = {}^\omega \omega$ as a metric space as follows:

For any $x, y \in \mathbb{R}, x \neq y$, let $n < \omega$ be the least such that $x(n) \neq y(n)$ then set $d(x, y) = \frac{1}{n+1}$.

For $s \in {}^{<\omega} \omega$, define $U_s = \{x \in {}^\omega \omega : s \subset x\}$ and recursively define $\mu(U_s)$ as follows: let $\mu(U_{\{\emptyset\}}) = 1$, now suppose $\mu(U_s)$ is already defined, then let $\mu(U_{s \smallfrown n}) = \frac{1}{2^{n+1}} \cdot \mu(U_s)$. Note that U_s is the disjoint union of $\{U_{s \smallfrown n} : n < \omega\}$, we call $U_s, s \in {}^{<\omega} \omega$ the collection of basic open sets.

A set $\mathcal{O} \subseteq {}^\omega \omega$ is **open** if and only if $\mathcal{O} = \bigcup_{s \in X} U_s$ for some $X \subseteq {}^{<\omega} \omega$ (possibly $X = \emptyset$) where $\forall s_1, s_2 \in X (s_1 \neq s_2 \rightarrow U_{s_1} \cap U_{s_2} = \emptyset)$. With this notion, ${}^\omega \omega$ is a topological space which in fact is the topology derived from the metric we defined earlier. And we may then set $\mu(\mathcal{O}) = \sum_{s \in X} \mu(U_s)$.

A set $N \subseteq {}^\omega \omega$ is called **null** if and only if $\forall \varepsilon > 0, \exists \mathcal{O} \subseteq {}^\omega \omega$ open, $\mathcal{O} \supseteq N, \mu(\mathcal{O}) \leq \varepsilon$.

A set $M \subseteq {}^\omega \omega$ is called **Lebesgue measurable** if and only if $\exists \mathcal{O} \subseteq {}^\omega \omega$ open such that $M \Delta \mathcal{O}$ is null.

A set $N \subseteq {}^\omega \omega$ is called **nowhere dense** if and only if for any $s \in {}^{<\omega} \omega$, there exists some $t \in {}^{<\omega} \omega, t \supset s$ such that $N \cap U_t = \emptyset$.

A set $M \subseteq {}^\omega \omega$ is called **meager** if and only if $M = \bigcup_{n < \omega} N_n$ for a collection of nowhere dense sets.

We say a set $A \subseteq {}^\omega \omega$ has the **Baire property** if and only if $\exists \mathcal{O} \subseteq {}^\omega \omega$ open such that $A \Delta \mathcal{O}$ is meager.

We say $I \subseteq \mathcal{P}({}^\omega \omega)$ is a **σ -ideal** if and only if $(\forall A \in I)(\forall B \subset {}^\omega \omega)(B \subseteq A \rightarrow B \in I)$ and for any $\{A_n : n < \omega\} \subset I, \bigcup_{n < \omega} A_n \in I$. Note that both the collections of null sets and meager sets form a σ -ideal.

A set $B \subseteq {}^\omega \omega$ is called **Borel** if $B \in \Gamma$ where $\Gamma \subset \mathcal{P}({}^\omega \omega)$ is the smallest class such that all open subset of ${}^\omega \omega$ in Γ and close under taking complement and countable union.

A set $P \subseteq {}^\omega \omega$ is called **perfect** if and only if $P \neq \emptyset, P$ is closed and $\forall x \in P, x$ is an accumulation point of P , i.e., $\forall n < \omega, (U_{x \restriction_n} \setminus \{x\}) \cap P \neq \emptyset$.

We say a set $A \subseteq {}^\omega \omega$ has the **perfect set property** if and only if either A is countable or A has a perfect subset.

Lemma 2.35. Let $\mathcal{L}$ be the set of all Lebesgue measurable subsets of ${}^\omega\omega$. Let $\mathcal{C}$ be the set of all $B \subseteq {}^\omega\omega$ such that there is some open set $A \subseteq {}^\omega\omega$ with $B \triangle A$ being meager. Both $\mathcal{L}$ and $\mathcal{C}$ form a σ -algebra containing all the open sets.

Proof. See Sheet 8 Exercise 2. ■

Corollary 2.36. Every Borel set is Lebesgue measurable and has the Baire Property.

Let $I \subseteq \mathcal{P}({}^\omega\omega)$ be a σ -ideal, then we consider the following poset: let $\mathcal{B}$ be the collection of all Borel sets, now we define an equivalence relation $\sim$ on $\mathcal{B}$ as follows: for $B_1, B_2 \in \mathcal{B}$, $B_1 \sim B_2 \iff B_1 \triangle B_2 \in I$. Now let us check if $\sim$ is an equivalence relation, for reflexivity and symmetry it is clear, as for transitivity, let $B_1 \sim B_2, B_2 \sim B_3$ then $B_1 \triangle B_2 \in I, B_2 \triangle B_3 \in I$, since $B_1 \setminus B_3 \subseteq (B_1 \setminus B_2) \cup (B_2 \setminus B_3)$ and $B_3 \setminus B_1 \subseteq (B_3 \setminus B_2) \cup (B_2 \setminus B_1)$. Thus $B_1 \triangle B_3 \subseteq (B_1 \triangle B_2) \cup (B_2 \triangle B_3)$, so $B_1 \triangle B_3 \in I$, i.e., $B_1 \sim B_3$. Hence $\sim$ is an equivalence relation on $\mathcal{B}$. Let $[B] = \{B' \in \mathcal{B} : B \sim B'\}$ the equivalence class of $B \in \mathcal{B}$, we say $[B] \leq [C]$ if and only if $B \setminus C \in I$.

We write $\mathcal{B}/I$ for the collection of all equivalence classes $[B], B \notin I$ and ordered by $\leq$. This order is well-defined because: Pick $\bar{B} \in [B], \bar{C} \in [C]$, suppose $\bar{B} \setminus \bar{C} \in I$ then $B \setminus C \subseteq (B \setminus \bar{B}) \cup (\bar{B} \setminus \bar{C}) \cup (\bar{C} \setminus C) \in I$ thus $B \setminus C \in I$ so $[B] \leq [C]$.

We write $\mathcal{B}/I$ for the collection of all equivalence classes $[B]$ where $B \notin I$ and ordered by $\leq$.

Lemma 2.37. There is a dense embedding j from Cohen forcing $\mathbb{C}$ to $\mathcal{B}/\text{meager}$.

Proof. Let $j(s) = [U_s]$ where $s \in {}^{<\omega}\omega = \mathbb{C}$, j is clearly an embedding. As for density, let $[B] \in \mathcal{B}/\text{meager}$, then by definition B is not meager and since B is Borel so B has the property of Baire, thus there is a non-empty open set $\mathcal{O}$ with $\mathcal{O} \triangle B$ is meager. Let $U_s \subseteq \mathcal{O}$ then $U_s \setminus B$ is meager thus $[U_s] \leq [B]$. ■

Lemma 2.38. Both $\mathcal{B}/\text{null}$ and $\mathcal{B}/\text{meager}$ have the c.c.c.

Proof. For $\mathcal{B}/\text{meager}$, since $\mathbb{C}$ has the c.c.c. and there exists a dense embedding $j: \mathbb{C} \rightarrow \mathcal{B}/\text{meager}$, it is easy to check that dense embedding preserves κ -c.c. thus $\mathcal{B}/\text{meager}$ has the c.c.c.

Let $\{[B_i] : i < \omega_1\}$ be an antichain in $\mathcal{B}/\text{null}$. If $[B_1], [B_2]$ are incompatible, consider $B_1 \cap B_2$, if $B_1 \cap B_2$ is not null then $[B_1 \cap B_2] \leq [B_1], [B_2]$ which makes $[B_1], [B_2]$ compatible. Thus $B_1 \cap B_2$ must be null. Hence $B_i \cap B_j$ is null for all $i \neq j$. As none of the B_i is null, $B_i^* = B_i \setminus \bigcup_{j < i} B_j$ is not null either. But now $B_i^* \cap B_j^* = \emptyset$ for $i \neq j$. We now have for each $i < \omega_1$, there is $n < \omega$ with $\mu(B_i^*) \geq \frac{1}{n+1}$. Then there is some $n < \omega$ and some $X \subseteq \omega_1$ of size $\aleph_1$ with $\mu(B_i^*) \geq \frac{1}{n+1}$ for all $i \in X$ which leads to a contradiction (that $\mu({}^\omega\omega) = 1$). ■

Now we look at Borel sets: $\Sigma_1^0 =$ open subsets of $\mathbb{R}, \Pi_1^0 =$ closed subsets of $\mathbb{R}, \Sigma_2^0 = F_\sigma =$ all countable union of closed subsets of $\mathbb{R}, \Pi_2^0 = G_\delta =$ all countable intersection of open subsets of $\mathbb{R}$.

Suppose $\alpha < \omega_1$ and all $\Sigma_\beta^0, \Pi_\beta^0$ are defined for $\beta < \alpha$ then:

$\Sigma_\alpha^0 =$ all countable union of elements of $\bigcup_{\beta < \alpha} \Pi_\beta^0, \Pi_\alpha^0 =$ all complements of elements of $\Sigma_\alpha^0 =$ all countable intersection of elements of $\bigcup_{\beta < \alpha} \Sigma_\beta^0$.

It is clear that $\mathcal{B} =$ the collection of Borel sets $= \bigcup_{\alpha < \omega_1} \Sigma_\alpha^0 = \bigcup_{\alpha < \omega_1} \Pi_\alpha^0$.

Now let M be a transitive model of ZFC and let g be M -generic. By recursion on $\alpha < \omega_1^M$ we define $B \mapsto B^* = B_g$ for $B \in (\Sigma_\alpha^0)^M$. If B is open in M , say $M \models B = \bigcup_{s \in X} U_s$ for some $X \subset {}^{<\omega}\omega$ then B^* is such that $M[g] \models B^* = \bigcup_{s \in X} U_s$.

Note that in $M, U_s = \{x \in {}^\omega\omega \cap M : x \supset s\}$; in $M[g], U_s = \{x \in {}^\omega\omega \cap M[g] : x \supset s\}$ since the real numbers in these two sets are different, so the basic open set is different.

Lemma 2.39. *Let M be a transitive model of ZFC, let g be $(\mathcal{B}/\text{null})^M$ -generic filter over M . Then there is a unique $x = x_g \in {}^\omega\omega \cap M[g]$ such that for all $B \in \mathcal{B}^M, x \in B^* \iff [B] \in g$.*

Proof. Existence: We work in $M[g]$ to construct $(S_n : n < \omega)$ where $S_n \in {}^{<\omega}\omega, \text{dom}(S_n) = n$ such that $[U_{S_n}^M] \in g$ for all $n < \omega$. Let $S_0 = \emptyset, U_\emptyset^M = [{}^\omega\omega \cap M] = \mathbb{1}_{(\mathcal{B}/\text{null})^M} \in g$. Now suppose S_n is defined, let $\mathcal{D} = \{[B] \in (\mathcal{B}/\text{null})^M : \exists k < \omega, [B] \leq [U_{S_n \smallfrown k}^M]\}$ then $\mathcal{D} \in M$.

We claim that $\mathcal{D}$ is dense below $[U_{S_n}^M]$ (here actually $[U_{S_n}^M] \in g$ by our construction). Let $[B] \leq [U_{S_n}]$, since $\bigsqcup_{k < \omega} B \cap U_{S_n \smallfrown k}^M = [B]$ and B is not null, the countable union of null sets is still null, thus $\exists k < \omega$ such that $B \cap U_{S_n \smallfrown k}^M$ is not null in M . For such $k < \omega, [B \cap U_{S_n \smallfrown k}^M] \leq [B]$, and thus $\mathcal{D}$ is dense below $[U_{S_n}]$. Then $g \cap \mathcal{D} \neq \emptyset$, say $[B] \in g \cap \mathcal{D}$, then set $S_{n+1} = S_n \smallfrown k$ for such k .

And we then set $x = \bigcup S_n$, now we verify the property that for all $B \in \mathcal{B}^M, x \in B^* \iff [B] \in g$ by recursion on Borel hierarchy. It is clear when $B = U_s$ for basic open sets $U_s, s \in {}^{<\omega}\omega$. Suppose $x \in B^* \iff [B] \in g$, then $x \notin B^* \iff x \in ({}^\omega\omega^M \setminus B)^* \iff [B] \notin g \iff [{}^\omega\omega^M \setminus B] \in g$. Suppose $(B_n : n < \omega) \in M, B_n \in \mathcal{B}^M$ and for all $n < \omega, x \in B_n^* \iff [B_n] \in g$. We want to show $x \in (\bigcap B_n)^* = \bigcap B_n^* \Rightarrow [\bigcap B_n] \in g$.

Since $[B_n] \in g$ and $[B_0 \cap B_1 \cap \dots \cap B_n] \in g$ for all $n < \omega$. Look at $\bigcap B_n, {}^\omega\omega^M \setminus B_1, {}^\omega\omega^M \setminus B_2, \dots$. If $[D] \in g, D = \bigsqcup_{n < \omega} D_n$ then at least one $[D_n] \in g$ because $\{[D_n] : n < \omega\} \cup \{[{}^\omega\omega^M \setminus D]\}$ is an maximal antichain in $(\mathcal{B}/\text{null})^M$. Let $[B] \leq [D], B \cap D$ is not null then one of $B \cap D_n$ is not null. However none of $[{}^\omega\omega^M \setminus B_i]$ can be in g . Hence $\bigcap B_n$ must lie in g .

For the other direction, $[\bigcap_{n < \omega} B_n] \in g \implies [B_n] \in g$ for all $n < \omega$ so $x \in B_n^*$ for all $n < \omega$.

Uniqueness: Let $x, x' \in {}^\omega\omega \cap M[g]$ be such that $\forall B \in \mathcal{B}^M, x \in B^* \iff [B] \in g \iff x' \in B^*$. Then for all $n < \omega, U_{x|_n}^M \subseteq B$ with $x \in (U_{x|_n}^M)^*$ thus $[U_{x|_n}^M] \in g$ and $x' \in (U_{x|_n}^M)^*$. Hence $x = x'$. $\blacksquare$

The unique real x_g as above for $(\mathcal{B}/\text{null})^M$ is called a **random real** over M .

The same holds for $\mathcal{B}/\text{meager}$, and the unique real for $(\mathcal{B}/\text{meager})^M$ is called a Cohen real over M .

Lemma 2.40. *Let M be a transitive model of ZFC and let $x \in {}^\omega\omega$ be in a generic extension of M . TFAE:*

- (1) $x \in {}^\omega\omega$ is a random (Cohen) real over M .
- (2) $x \notin B^*$ for all $B \in \mathcal{B}^M$ such that $M \models B$ is null (meager).

Proof. (1) $\implies$ (2): Let g be a $(\mathcal{B}/\text{null})^M$ -generic over M witness x is a random real. Let B be a null set in M . Since ${}^\omega\omega^M \setminus B \in [{}^\omega\omega^M] \in g$ so $x \in ({}^\omega\omega^M \setminus B)^* = {}^\omega\omega^{M[g]} \setminus B^*$ then $x \notin B^*$.

(2) $\implies$ (1): Let x satisfies (2). Let $g := \{[B] : B \in \mathcal{B}^M \wedge x \in B^*\}$. It suffices to show g is $(\mathcal{B}/\text{null})^M$ -generic over M . Let $[B] \in g$ and $[C] \geq [B]$. Since $B \setminus C$ is null, by (2), $x \notin (B \setminus C)^*$ thus $x \in (B \cap C)^*, [C] \in g$. Let $[B_1], [B_2] \in g$ since $x \in B_1^* \cap B_2^*$, then $x \in U_s^{M[g]} \subseteq B_1^* \cap B_2^* = (B_1 \cap B_2)^*$ and

$g \ni [B_1 \cap B_2] \leq [B_1], [B_2]$. Now let $A \subseteq (\mathcal{B}/\text{null})^M$ be an maximal antichain, $A \in M, M \models A$ is countable. So $A = \{[B_n] : n < \omega\} \in M$. Therefore ${}^\omega\omega \setminus \bigcup B_n$ is null, otherwise the antichain would not be maximal. Hence $x \in \bigcup B_n^*$ so $\exists n < \omega, x \in B_n^*$. Thus $g \cap A \neq \emptyset$. ■

Lemma 2.41. *Let M be a transitive model of ZFC and g be a M -generic then in $M[g]$:*

If $\mathbb{R} \cap M$ is countable in $M[g]$ then

- (1) $A = \{x \in {}^\omega\omega : x \text{ is not random over } M\}$ is a null set.
- (2) $A = \{x \in {}^\omega\omega : x \text{ is not Cohen over } M\}$ is a meager set.

Proof. We show (1) as the similar argument gives (2).

By Lemma 2.40, $x \in A \iff x \in B^*$ for some $B \in M$ such that $M \models "B \text{ is a Borel null set}"$. If $M \models B$ is Borel null, then $M[g] \models B^*$ is Borel null (**Why?** Because the Lebesgue measure is absolute between M and $M[g]$). Therefore $A = \bigcup \{B^* : B \in M \wedge M \models B \text{ is Borel and null}\}$. Since $M \models$ there are $2^{\aleph_0}$ many Borel sets as there are $2^{\aleph_0}$ many open subsets. Hence in $M[g]$, A is a countable union of null sets thus still null. ■

Definition 2.42. *Let M be a transitive model of ZFC and g be M -generic. A set $A \in M[g]$ such that $M[g] \models "A \text{ is a set of reals}"$ is called **Solovay over M** if and only if there is a formula φ and there are parameters $x_1, \dots, x_k \in M$ such that for all $z \in {}^\omega\omega \cap M[g]$ which are generic over M ,*

$$z \in A \iff M[z] \models \varphi(z, x_1, \dots, x_k).$$

Remark: Here we may define z is generic over M as: z is a random real or a Cohen real. This suffices for our purpose. It is not hard to see that every Borel set B^* where $B \in M$ is Solovay over M . We will eventually prove that every set of reals in M is Solovay over some $M[g \restriction \xi], \xi < \kappa$.

Lemma 2.43. *Let M be a transitive model of ZFC and g be a M -generic.*

Let $A \in M[g]$, a subset of ${}^\omega\omega \cap M[g]$, be Solovay over M . Then

- (1) *there is some $B \in M[g]$, B a Borel set in $M[g]$ such that $A \triangle B \subset \{x \in {}^\omega\omega \cap M[g] : x \text{ is not random over } M\}$.*
- (2) *there is some $B' \in M[g]$, B' a Borel set in $M[g]$ such that $A \triangle B' \subset \{x \in {}^\omega\omega \cap M[g] : x \text{ is not Cohen over } M\}$.*

Proof. We show (1) as the very similar argument gives (2).

By assumption, there is a formula φ and some elements $x_1, \dots, x_k \in M$ such that $z \in A \iff M[z] \models \varphi(x_1, \dots, x_k)$ for any z which is generic over M . It suffices to verify the property above for reals z which are random over M . Let τ be a canonical name in $M^{(\mathcal{B}/\text{null})^M}$ for a random real over M . Let $E \subset (\mathcal{B}/\text{null})^M, E \in M$ be an maximal antichain in M consist of $[D]$ with $[D] \Vdash_M^{(\mathcal{B}/\text{null})^M} \varphi(\tau, \check{x}_1, \dots, \check{x}_k)$. Since $\mathcal{B}/\text{null}$ has the c.c.c. so $\overline{\overline{E}} \leq \aleph_0$ in M . Write $E = \{[D_n] : n < \omega\}$ in M . Set $B = \bigcup_{n < \omega} D_n^* = \left(\bigcup_{n < \omega} D_n \right)^*$, then B is Borel in $M[g]$. We claim that if $z \in M[g]$ is random over M , then $z \in A \iff z \in B$. Now let $z \in M[g]$ be random over M . Then $z \in A \iff M[z] \models \varphi(z, x_1, \dots, x_k) \iff \exists [D] \in h \cap E, [D] \Vdash_M^{(\mathcal{B}/\text{null})^M} \varphi(\tau, \check{x}_1, \dots, \check{x}_k) \iff \exists n < \omega, [D_n] \in h \iff [\bigcup D_n] \in h \iff z \in B$.

*: Since z is random so $z = \tau^h$ for some $(\mathcal{B}/\text{null})^M$ -generic h , $M[z] = M[\tau^h]$. ■

Corollary 2.44. *Let $M, g, M[g]$ be as before such that $\mathbb{R} \cap M$ is countable in $M[g]$. If $A \in M[g]$, a set of reals in $M[g]$, is Solovay over M . Then A is lebesgue measurable and has the Baire property.*

Proof. Follows immediately by the two previous lemma. ■

Definition 2.45. *A forcing poset $\mathbb{P}$ is called separative if and only if $\forall p \in \mathbb{P}, \forall q \in \mathbb{P} (q \not\leq p \rightarrow (\exists r \in \mathbb{P}, r \leq q \wedge r \perp p))$.*

Lemma 2.46. *Let κ be an infinite cardinal and $\mathbb{P}$ be an atomless forcing poset such that $\mathbb{1}_{\mathbb{P}} \Vdash \bar{\kappa} = \check{\aleph}_0$. Then for every $p \in \mathbb{P}$, there is an antichain $A \subseteq \{q \in \mathbb{P} : q \leq p\}$ of size κ .*

Proof. Fix $p \in \mathbb{P}$. First, assume that $\kappa = \omega$. Now we construct two sequences of conditions $(p_n : n < \omega)$ and $(q_n : n < \omega)$ as follows: Let $p_0 = p$, suppose p_n is given then let p_{n+1}, q_n be two incompatible extension of p_n then $(q_n : n < \omega)$ is the desired sequence.

Now assume $\text{cf}(\kappa) = \omega < \kappa$, let $(\kappa_n : n < \omega)$ be a sequence of uncountable regular cardinals which is cofinal of κ . Then construct a sequence of conditions $(p_n : n < \omega)$ and a sequence of antichains in $\mathbb{P}$ $(A_n : n < \omega)$ as follows:

Set $p_0 = p$, suppose p_n is given, notice that $p_n \Vdash \bar{\kappa} = \check{\aleph}_0$ by lemma 6.32 in Schindler's book, there exists an antichain $A \subseteq \{p \in \mathbb{P} : p \in p_n\}$ of size κ_n . This is because if $\mathbb{P}$ has the κ_n -c.c. then κ_n will be a cardinal in the generic model as it is uncountable regular, but since $\mathbb{1}_{\mathbb{P}} \Vdash \bar{\kappa} = \check{\aleph}_0$, which means κ_n is no longer a cardinal in the generic model, contradiction. Thus there must be an antichain with size of κ_n in $\{q \in \mathbb{P} : q \leq p\}$. Let A_n be such an antichain and let $p_{n+1} \in A_n$ be arbitrary. Then $\bigcup \{A_n \setminus \{p_{n+1}\} : n < \omega\} \subset \{q \in \mathbb{P} : q \leq p\}$ is an antichain of size κ .

Finally we assume $\text{cf}(\kappa) > \omega$. Let $r \leq p, \tau \in M^{\mathbb{P}}$ such that $r \Vdash \check{\omega} \rightarrow \check{\kappa}$ is surjective. Let us suppose that every antichain in $\{q \in \mathbb{P} : q \leq r\}$ have size smaller than κ . For every $n < \omega$, let A_n be an maximal antichain in $\{q \in \mathbb{P} : q \leq r \wedge \exists \xi q \Vdash \tau(\check{n}) = \check{\xi}\}$. By our hypothesis, $\overline{A_n} < \kappa$ for all $n < \omega$ so by $\text{cf}(\kappa) > \omega$, there can not be a surjective function $f : (\omega \times \bigcup \{A_n : n < \omega\}) \rightarrow \kappa$.

However we now construct such an f by setting, for $n < \omega$ and $p \in \bigcup \{A_n : n < \omega\}$,

$$f((n, p)) = \begin{cases} \xi & \text{if } p \Vdash \tau(\check{n}) = \check{\xi} \\ 0 & \text{if no } \xi < \kappa \text{ such that } p \Vdash \tau(\check{n}) = \check{\xi} \end{cases}$$

If $\xi < \kappa$ then there is some $r' \leq r, n < \omega$ such that $r' \Vdash \tau(\check{n}) = \check{\xi}$. But then $q \parallel r'$ for some $q \in A_n$ by the choice of A_n , so that $q \Vdash \tau(\check{n}) = \check{\xi}$. Therefore f is surjective, which leads to a contradiction. ■

Lemma 2.47. *Let $\mathbb{P}$ be a separative forcing poset and $\mu \geq \aleph_1$ be a cardinal, if $\bar{\mathbb{P}} = \mu$ and $\mathbb{1}_{\mathbb{P}} \Vdash \check{\mu}$ is countable" then there is a dense embedding $e : \text{Col}(\omega, \mu) \rightarrow \mathbb{P}$.*

Proof. Let g be a $\mathbb{P}$ -generic filter over M and τ be such that $\mathbb{1}_{\mathbb{P}} \Vdash \check{\omega} \rightarrow \check{\mu}$ is onto". We construct e by recursion on the size of domain. First set $e(\mathbb{1}_{\text{Col}(\omega, \mu)}) = \mathbb{1}_{\mathbb{P}}$. Suppose $p \in \text{Col}(\omega, \mu)$, $\text{dom}(p) = n$ and $e(p)$ was already chosen. Since $\mathbb{1}_{\mathbb{P}} \Vdash \bar{\mu} \leq \aleph_0$ and $\bar{\mathbb{P}} = \mu$, by 2.46, we may pick some maximal antichain $A \subseteq \mathbb{P}$ of conditions with $\bar{A} = \mu, q \leq e(p)$ such that q decides $\tau(\check{n})$, i.e., there exist some $r \in \mathbb{P}$ with $q \Vdash \tau(\check{n}) = \check{r}$. We write $A = \{q_\xi : \xi < \mu\}$, then define $e(p \cup \{(n, \xi)\}) = q_\xi$ for $\xi < \mu$.

Thus for each $n < \omega$, $A_n = \{e(p) : p \in \text{Col}(\omega, \mu), \text{dom}(p) = n\}$ is an maximal antichain of conditions $q \in \mathbb{P}$ such that q decides $\tau(\check{0}), \dots, \tau(\check{n} - 1)$.

To see that e is dense, let $r \in \mathbb{P}, r \Vdash \check{r} \in \dot{g}$. Hence there are some $s \leq r$ such that $s \Vdash \tau(\check{n}) = \check{r}$ for some $n < \omega$ (**By density argument**). Let $t \leq s$ be such that t decides $\tau(\check{0}), \dots, \tau(\check{n})$, then there is some $p \in \text{Col}(\omega, \mu)$ such that $e(p) \in A_{n+1}, e(p) \parallel t$. Then we have $e(p) \Vdash \tau(\check{n}) = \check{r}$. Hence $e(p) \Vdash \check{r} \in \dot{g}$. As $\mathbb{P}$ is separative, this gives $e(p) \leq r$, otherwise then $\exists u \leq e(p), u \perp r$ but then $u \Vdash \check{r} \in \dot{g}$ and $u \parallel r$ which leads to a contradiction. $\blacksquare$

2.2.1 The BIG Theorem and its LONG Proof

Theorem 2.48 (Part 1). *Let κ be inaccessible and $G \subseteq \text{Col}(\omega, < \kappa)$ be M -generic. Then in $M[G]$, every set of reals which is $\text{OD}_{\omega_\omega}$ is Lebesgue measurable and has the Baire property.*

Recall: By Lemma 2.34, every $x \in H_{\omega_1^{M[G]}}^M[G]$ appears in $M[G \restriction_\lambda]$ for some $\lambda < \kappa$.

Now we begin to proof this theorem.

Claim 2.49. *Let $\lambda < \kappa, \mathbb{P} \in H_{\kappa}^{M[G]}$ (since κ collapse to ω_1 in $M[G]$ so $H_{\kappa}^{M[G]} \subseteq H_{\omega_1^{M[G]}}^{M[G]}$). Let $S \in M[G]$ be $\mathbb{P}$ -generic over $M[G \restriction_\lambda]$. Then there is $H \subseteq \text{Col}(\omega, < \kappa)$ which is generic over $M[G \restriction_\lambda][S]$ such that $M[G] = M[G \restriction_\lambda][S][H]$.*

Why we only consider the $\text{OD}_{\omega_\omega}$ sets?

What will happen if this is true? Let $A \subseteq {}^\omega\omega$ in $M[G]$ be $\text{OD}_{\omega_\omega}$, then there is a formula φ , some reals $\vec{x}$ and some ordinals $\vec{\alpha}$ such that for any real $x, x \in A \iff M[G] \models \varphi(x, \vec{x}, \vec{\alpha})$. Let $\lambda < \kappa$ be such that $\vec{x} \in M[G \restriction_\lambda]$. By Corollary 2.44, to show A is Lebesgue measurable and has the Baire property, we need to show 2 things: First, $(2^{\aleph_0})^{M[G]}$ is countable which holds true trivially. Second, A is Solovay over $M[G \restriction_\lambda]$. To show this, let $x \in M[G] \cap {}^\omega\omega$ be generic over $M[G \restriction_\lambda], x \in A \iff M[G] \models \varphi(x, \vec{x}, \vec{\alpha})$, by Claim 2.49, $\iff M[G \restriction_\lambda][x][H] \models \varphi(x, \vec{x}, \vec{\alpha}) \xLeftrightarrow{\text{by FT}} \exists p \in H, p \Vdash_{\text{Col}(\omega, < \lambda)}^{M[G \restriction_\lambda][x]} \varphi(\dot{x}, \vec{x}, \vec{\alpha})$, by homogeneity, $\iff \mathbb{1}_{\text{Col}(\omega, < \lambda)} \Vdash \varphi(\dot{x}, \vec{x}, \vec{\alpha})$, it is equivalent to $M[G \restriction_\lambda][x] \models \varphi(x, \vec{x}, \vec{\alpha})$, so A is Solovay over $M[G \restriction_\lambda]$.

Proof. For notation simplicity, we may assume $\lambda = 0$. By lemma 2.34, there exists some $\mu + 1 < \kappa$ such that $S \in M[G \restriction_{\mu+1}]$. Note that $\overline{\text{Col}(\omega, < \mu + 1)} = \bar{\mu}$ in M and also $\mathbb{1}_{\text{Col}(\omega, < \mu+1)} \Vdash \check{\mu}$ is countable. By lemma 2.47, there is a dense embedding $i: \text{Col}(\omega, \mu) \rightarrow \text{Col}(\omega, < \mu + 1)$.

Let $G_0 = \{p \in \text{Col}(\omega, \mu) : i(p) \in G \restriction_{\mu+1}\}, M[G_0] = M[G \restriction_{\mu+1}]$. Also $j: \text{Col}(\omega, \mu) \cong \text{Col}(\omega, \{\mu + 1\})$, let $G_1 = \{p \in \text{Col}(\omega, \mu) : \exists q \in G, q(\mu + 1) = j(p)\}$. Then G_1 is $\text{Col}(\omega, \mu)$ -generic over $M[G_0] = M[G \restriction_{\mu+1}]$ such that $M[G_0][G_1] = M[G \restriction_{\mu+2}]$.

Claim 2.50. *There is some $\text{Col}(\omega, \mu)$ -generic filter H^* over $M[S]$ with $M[S][H^*] = M[G_0][G_1]$.*

What will happen if this is true? We will then have $M[G] = M[G \restriction_{\mu+2}][G \restriction_{[\mu+2, \kappa)}] = M[G_0][G_1][G \restriction_{[\mu+2, \kappa)}] = M[S][H^*][G \restriction_{[\mu+2, \kappa)}]$. Let $i': \text{Col}(\omega, \mu) \rightarrow \text{Col}(\omega, < \mu + 2)$ be a dense embedding. Let $H_0 = \{p \in \text{Col}(\omega, < \mu + 2) : \exists q \in H^*, i'(q) \leq p\}$. Then H_0 is $M[S]$ -generic with $M[S][H_0] = M[S][H^*]$. Finally set $H = \{p \in \text{Col}(\omega, < \kappa) : p \restriction_{\mu+2} \in H_0 \wedge p \restriction_{[\mu+2, \kappa)} \in G \restriction_{[\mu+2, \kappa)}\}$, then H is $\text{Col}(\omega, < \kappa)$ -generic over $M[S]$ and $M[S][H] = M[S][H_0][G \restriction_{[\mu+2, \kappa)}] = M[S][H^*][G \restriction_{[\mu+2, \kappa)}] = M[G]$. And this gives claim 2.49.

Proof. As $S \in M[G_0]$, there is a $\text{Col}(\omega, \mu)$ -name $\tau \in M$ such that $S = \tau^{G_0}$.

Recursively define $(\mathbb{Q}_\alpha : \alpha \in \text{Ord})$ of subsets of $\text{Col}(\omega, \mu)$. Set $p \in \mathbb{Q}_0$ if and only if

$$\forall r \in \mathbb{P}, \begin{cases} p \Vdash \check{r} \in \tau \implies r \in S \\ p \Vdash \check{r} \notin \tau \implies r \notin S \end{cases}$$

For $\alpha + 1, p \in \mathbb{Q}_{\alpha+1} \iff$ for any open dense set $D \subseteq \text{Col}(\omega, \mu), D \in M, \exists p' \leq p: p' \in D \cap \mathbb{Q}_\alpha$.

For α is limit, $\mathbb{Q}_\alpha = \bigcap_{\beta < \alpha} \mathbb{Q}_\beta$. Note that $\mathbb{Q}_\alpha$ s are upwards closed. And $\alpha < \beta \implies \mathbb{Q}_\beta \subseteq \mathbb{Q}_\alpha$ because if $p \in \mathbb{Q}_{\alpha+1}$ then $\exists p' \leq p, p' \in D \cap \mathbb{Q}_\alpha$ so $p \in \mathbb{Q}_\alpha$.

It must be the case that $\exists s < \text{Ord}$ such that $\mathbb{Q}_\delta = \mathbb{Q}_{\delta+1}$, i.e., the sequence will become stationary at some point and let δ be the least such. Denote $\overline{\mathbb{Q}} = \mathbb{Q}_\delta, \mathbb{Q} = \overline{\mathbb{Q}} \times \text{Col}(\omega, \mu)$.

We construe $\overline{\mathbb{Q}}$ and $\mathbb{Q}$ as partial orders, with the order relation given by the restriction of the order relation of $\text{Col}(\omega, \mu)$ and $\text{Col}(\omega, \mu) \times \text{Col}(\omega, \mu)$ to $\overline{\mathbb{Q}}$ and $\mathbb{Q}$ respectively. If $p \in \overline{\mathbb{Q}}$ and $p \leq q, q \in \text{Col}(\omega, \mu)$ then $q \in \overline{\mathbb{Q}}$.

Notice that $\overline{\mathbb{Q}}$ was defined inside $M[S]$ with the parameters $\mu, \mathbb{P}, \tau, S$. We write $\Psi(\nu_0, \nu_1, \nu_2, \nu_3, \nu_4)$ for the defining formula, i.e., $M[S] \models \forall \mathbb{Q}' (\mathbb{Q}' = \overline{\mathbb{Q}} \leftrightarrow \Psi(\mathbb{Q}', \mu, \mathbb{P}, \tau, S))$.

Subclaim 2.51. $G_0 \subseteq \overline{\mathbb{Q}}$.

Proof. Suppose $p \in G_0 \setminus \overline{\mathbb{Q}}$ where β is the least such $G_0 \setminus \overline{\mathbb{Q}}_\beta \neq \emptyset$. By the construction of $\overline{\mathbb{Q}}$, β can not be 0 or a limit ordinal, thus $\beta = \alpha + 1$ for some α . Since $p \notin \overline{\mathbb{Q}}_\beta$ so there exists an open dense $D \subseteq \text{Col}(\omega, \mu), D \in M$ such that $\forall p' \leq p, p' \notin D \cap \mathbb{Q}_\alpha$. Notice G_0 is generic so $G_0 \cap D \neq \emptyset$, say $p^* \in G_0 \cap D$. Then $p, p^* \in G_0$ there exist $r \in G_0, r \leq p, p^*$. But D is open so $r \in D \cap \mathbb{Q}_\alpha$ hence $G_0 \setminus \overline{\mathbb{Q}}_\alpha \neq \emptyset$, which contradicts the choice of β . $\blacksquare$

The subclaim above gives $\overline{\mathbb{Q}} \neq \emptyset$ also $\overline{\mathbb{Q}}$ is separative and has size μ .

Subclaim 2.52. Let $p \in \overline{\mathbb{Q}}$. Then in $M[G]$ there is some G'_0 $\text{Col}(\omega, \mu)$ -generic over M such that $p \in G'_0, S = \tau^{G'_0}$ where τ was chosen such that $\tau^{G_0} = S$.

Proof. Let $\bar{p} \in \overline{\mathbb{Q}}, D \subset \text{Col}(\omega, \mu), D \in M$ be open dense. There is some $p' \leq \bar{p}$ with $p' \in D \cap \mathbb{Q}_\delta$ by $\bar{p} \in \mathbb{Q}_{\delta+1} = \overline{\mathbb{Q}}$. Inside $M[G]$, there are only countably many dense subset of $\text{Col}(\omega, \mu)$ which are in M . Therefore inside $M[G]$, given any $p \in \overline{\mathbb{Q}}$, we may construct a $\text{Col}(\omega, \mu)$ -generic filter G'_0 over M with $p \in G'_0, G'_0 \subset \overline{\mathbb{Q}}$.

Let $r \in \mathbb{P}$ and suppose that $p' \in G'_0$ decides $\check{r} \in \tau$. As $p' \in G'_0 \subset \overline{\mathbb{Q}} \subset \mathbb{Q}_0$, hence $p' \Vdash \check{r} \in \tau \implies r \in S, p' \Vdash \check{r} \notin \tau \implies r \notin S$. Therefore G'_0 is as desired. $\blacksquare$

Subclaim 2.53. $G_0 \times G_1$ is $\mathbb{Q}$ -generic over $M[S]$.

Proof. Let $D \in M[S]$ be dense in $\mathbb{Q}$. We want to show $D \cap (G_0 \times G_1) \neq \emptyset$. Suppose $D \cap (G_0 \times G_1) = \emptyset$.

Let $\rho \in M^\mathbb{P}$ be such that $\rho^S = D$. Let $\Phi(\nu_0, \nu_1, \nu_2, \nu_3, \nu_4)$ be a formula such that if $G''_0 \times G''_1$ is $\text{Col}(\omega, \mu) \times \text{Col}(\omega, \mu)$ -generic over M , then $M[G''_0 \times G''_1] \models \Phi(G''_0, G''_1, \mu, \mathbb{P}, \tau)$ if and only if the following holds true: If $s' = \tau^{G''_0}$ then s' is $\mathbb{P}$ -generic over M and if $D' = \{(p, q) \in \text{Col}(\omega, \mu) \times \text{Col}(\omega, \mu) : \exists r \in$

$s' \wedge r \Vdash_M^{\mathbb{P}} (\check{p}, \check{q}) \in \rho\}$ and $M[S] \models \Psi(\mathbb{Q}', \mu, \mathbb{P}, \tau, S)$ is true for exactly one $\mathbb{Q}'$, then D is dense in $\mathbb{Q}'$ and $D' \cap (G_0'' \times G_1'') = \emptyset$.

By hypothesis $M[G_0 \times G_1] \models \Phi(G_0, G_1, \mu, \mathbb{P}, \tau)$, let $(p, q) \in G_0 \times G_1$ be such that

$$(p, q) \Vdash_M^{\text{Col}(\omega, \mu) \times \text{Col}(\omega, \mu)} \Phi(\dot{G}_0, \dot{G}_1, \check{\mu}, \check{\mathbb{P}}, \check{\tau})$$

where $\dot{G}_0, \dot{G}_1 \in M^{\text{Col}(\omega, \mu) \times \text{Col}(\omega, \mu)}$ are the canonical name for G_0, G_1 respectively. By subclaim 2.51, $(p, q) \in \mathbb{Q}$. Since D is dense in $\mathbb{Q}$, there is some $(p', q') \leq (p, q)$ such that $(p', q') \in D$. By subclaim 2.52, there is some $G_0' \times G_1'$ inside $M[G]$ which is $\text{Col}(\omega, \mu) \times \text{Col}(\omega, \mu)$ -generic over M such that $(p', q') \in G_0' \times G_1'$ and $\tau^{G_0'} = s$.

Now we have $M[G_0' \times G_1'] \models \Phi(G_0', G_1', \mu, \mathbb{P}, \tau)$ and $s' = \tau^{G_0'}$ and $D' = D$ for the s', D' which are mentioned in $\Phi(G_0', G_1', \mu, \mathbb{P}, \tau)$ also the $\mathbb{Q}' = \mathbb{Q}$. We then have $D \cap G_0' \times G_1' = \emptyset$.

However $(p', q') \in D \cap G_0' \times G_1'$. **CONTRADICTION!!** ■

Now by lemma 2.47, there is a dense embedding $k: \text{Col}(\omega, \mu) \rightarrow \mathbb{Q}$, set $H^* = \{p \in \text{Col}(\omega, \mu) : k(p) \in G_0 \times G_1\}$, then H^* is $\text{Col}(\omega, \mu)$ -generic over $M[S]$ and $M[S][H^*] = M[S][G_0 \times G_1] = M[G_0][G_1]$ as desired. ■

Now we finish the proof of inside $N = \text{HOD}_{M \cup (\omega \text{Ord} \cap M[G])}^{M[G]}$, every set of reals is Lebesgue measurable and has the Baire property. ■

Now let us show the perfect set property.

Theorem 2.54 (Part 2). *Let κ be inaccessible and $G \subseteq \text{Col}(\omega, < \kappa)$ be M -generic. Then in $M[G]$, every uncountable set of reals which is $\text{OD}_{\omega_\omega}$ has a perfect subset.*

Proof. Fix $A \subset N$ be an uncountable set of reals, then $x \in A \iff M[G] \models \varphi(x, z_0, \dots, z_k, u_0, \dots, u_l)$ where $z_0, \dots, z_k \in M, u_0, \dots, u_l \in M[G] \cap {}^\omega \omega$. We showed that there are some $\xi < \kappa$ such that $u_0, \dots, u_l \in M[G \restriction_\xi]$ and $\overline{{}^\omega \omega \cap M[G \restriction_\xi]} = \aleph_0$ in $M[G]$. Therefore $A \setminus M[G \restriction_\xi] \neq \emptyset$. Say $x \in A \setminus M[G \restriction_\xi]$, let $\eta > \xi$ be such that $x \in M[G \restriction_{(\eta+1)}]$. There is a dense embedding $e: \text{Col}(\omega, \eta) \rightarrow \text{Col}(\omega, [\xi, \eta])$, let $G^* = e^{-1} G \restriction_{[\xi, \eta]}$ then G^* is $\text{Col}(\omega, \eta)$ -generic over $M[G \restriction_\xi]$, $x \in M[G \restriction_\xi][G^*] \setminus M[G \restriction_\xi]$.

Let $\tau \in M[G \restriction_\xi]^{\text{Col}(\omega, \eta)}$ be such that $\tau^{G^*} = x$, let us construct $(p_s : s \in {}^{<\omega} 2)$ as follows:

Let $(D_n : n < \omega) \in M[G]$ be an enumeration of dense subsets of $\text{Col}(\omega, \eta)$ which exists in $M[G \restriction_\xi]$. Since $M[G] = M[G \restriction_\xi][G^*][G \restriction_{[\eta+1, \kappa)}]$ $M[G] \models \varphi(x, \vec{z}, \vec{u}) \iff \mathbb{1}_{\text{Col}(\omega, [\eta+1, \kappa))} \Vdash_{M[G \restriction_\xi][G^*]} \varphi(x, \vec{z}, \vec{u})$. This gives $G^* \ni p_\emptyset \Vdash \mathbb{1}_{\text{Col}(\omega, [\eta+1, \kappa))} \Vdash_{M[G \restriction_\xi][G^*]} \varphi(x, \vec{z}, \vec{u}) \wedge \tau \notin M[G \restriction_\xi]$ for some $p_\emptyset$. Now we fix such $p_\emptyset$ and assume $p_s \in \text{Col}(\omega, \eta)$ was already constructed, $p_s \leq p_\emptyset$. There must be some $n < \omega$ such that p_s does not decide $\tau(\check{n})$ as $p_\emptyset \Vdash \tau \notin M[G \restriction_\xi]$, otherwise $x \in M[G \restriction_\xi]$.

Choose $n < \omega, p_{s \smallfrown 0}, p_{s \smallfrown 1} \in \text{Col}(\omega, \eta)$ such that $p_{s \smallfrown 0} \leq p_s, p_{s \smallfrown 1} \leq p_s, p_{s \smallfrown 0} \Vdash \tau(\check{n}) = \check{m}, p_{s \smallfrown 1} \Vdash \tau(\check{n}) = \check{k}$ where $m \neq k$. Then $p_{s \smallfrown 0}, p_{s \smallfrown 1} \in D_{\text{dom}(s)}$, for every $z \in {}^\omega 2, \{q : \exists l, q \geq p_{z \restriction l}\}$ is $\text{Col}(\omega, \eta)$ -generic over $M[G \restriction_\xi]$, call it G_z^* .

Then $P := \{\tau^{G_z^*} : z \in {}^\omega 2 \cap M[G]\}$ is a perfect set in $M[G]$, because: Clearly it is not empty. We pick $z' \in {}^\omega 2 \cap M[G]$, if $z \restriction_n = z' \restriction_n$ then $\tau^{G_{z \restriction_n}^*} = \tau^{G_{z' \restriction_n}^*}$ thus every point is an accumulation point. Let $\tau^{G_{z_n}^*}$

convergence to y , which means $\forall k, \exists n(k), \forall n \geq n(k), z_n \upharpoonright_k = z_{n(k)} \upharpoonright_k$. Let $z = \bigcup_{k < \omega} z_{n(k)} \upharpoonright_k$ then $y = \tau^{G_z^*}$ this gives closedness.

By the choice of $\tau, P \subset A$. This gives the perfect set property. ■